

A Scenario-Based Evaluation of CRQC+AI Vulnerability Spectrum for TLS 1.3 Cryptographic Dependencies

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Abstract

This paper evaluates quantum and AI-accelerated risks to TLS 1.3 cryptographic dependencies under an evidence-tiered model, distinguishing mechanism-backed threats (Shor’s algorithm against RSA and ECC) from contingency-backed risks to lattice-based post-quantum cryptography (PQC) and hypothesis-only risks to hash-based and symmetric primitives. We do not identify any known breaks of ML-KEM, ML-DSA, SLH-DSA, or AES-256. Instead, we use explicit scenario assumptions, organized as a four-scenario capability model with parameters and pseudocode for reproducibility, to stress-test migration timelines across RSA-2048, ML-KEM (hybrid X25519 + Kyber-768 and standalone Kyber-1024), ML-DSA-44, SLH-DSA, and AES256, accompanied by one-at-a-time parameter sensitivity analysis and an explicit falsification analysis section. The primary methodological contribution is a reproducible scenario-estimation instrument together with its explicit update mechanics: every parameter is a named, anchored quantity that can be varied and the model rerun; a stated protocol maps observed conformance to, or deviation from, the modeled curves onto revisions of specific parameters, so progressive refinements can be tested against accumulating historical data. The paper is a methodological companion to quantum resource-estimation studies and to expert-elicitation timeline surveys such as the Global Risk Institute quantum threat reports, with its revision rules stated explicitly. As of mid-2026, the model does not show any NIST-approved algorithms as broken. Instead, the vulnerability spectrum under different scenarios shows RSA risk crossing the 50% threshold between 2030-2032 and the PQC risk becoming a non-zero risk after 2032-2035 under contingency scenarios conditional on the unproven dimension-collapse. The practical conclusions urge PQC migration be treated as mandatory by the 2030 and 2031 federal deadlines set by regulatory mandate and by Mosca’s HNDL reasoning, and that crypto-agility and hybrid cryptographic deployment be considered necessary complements to any PQC migration efforts.

Keywords: post-quantum cryptography; TLS 1.3; lattice-based cryptography; Learning with Errors; cryptographically relevant quantum computer; quantum machine learning; cryptanalysis; crypto-agility, quantum resource estimation; expert elicitation; scenario modeling; falsifiability

1. Introduction

In May 2021, the *Journal of Cryptology* published a special issue devoted to security of the Transport Layer Security (TLS) 1.3 protocol. [1] Since then, progress in the fields of quantum computing, quantum algorithms, and AI has exploded and heightened the need for NIST-standardized postquantum cryptography (PQC) to address the threats posed to classical public key cryptography. [2] In February 2024, the impact of Quantum-enhanced Artificial Intelligence (QAI), Quantum Machine Learning (QML) and Quantum Deep Learning (QDL) on the projected 2035 migration timeline for PQC encryption algorithms was explored in a Preprints paper (subsequently withdrawn over unresolved authorship concerns

unrelated to the substantive analysis presented in the paper). [3] These three classes of quantum-enhanced AI, and their relationship to the PQC migration timeline, are illustrated in Figure 1, reproduced here only for its uncontroversial depiction of the high-level interfaces and relationships of Quantum-accelerated AI/ML and DL in which ML and DL are subsets of AI and DL is a subset of ML.

Simplified Quantum-Classical Artificial Intelligence Interface

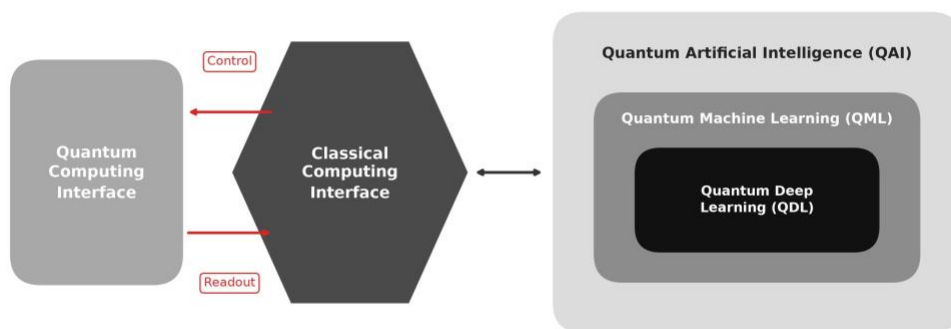


Figure 1: Simplified Quantum-Classical Artificial Intelligence Interface
(adapted from the withdrawn preprint [3]; used for conceptual framing only).

The National Institute of Standards and Technology (NIST) finalized its first PQC standards for asymmetric encryption algorithms of public-private key exchanges for key-encapsulation (FIPS 203) and digital signatures (FIPS 204 and 205) in August 2024. [4] But less than six months later, the federal timelines for implementing these PQC algorithms for certain classes of infrastructure systems were accelerated by executive order; and the most recent executive order as of June 2026 mandates that all federal agencies migrate all high value assets and high-impact systems (excluding National Security Systems) to PQC for key establishment by December 31, 2030, and for digital signatures by December 31, 2031. [5] Comparable updates to accelerate the PQC transition timelines have happened in the European Union and Australia. [6]



Figure 2: Federal PQC Transition Compliance Deadlines

The global efforts to implement PQC in various communication and identification protocols are underway. [7] Regardless of the actual transition deadlines for PQC, there is an open question as to whether implementation of PQC alone will be enough to protect TLS 1.3 from potential cryptographically relevant quantum computer (CRQC) and AI threats. These threats include not only the quantum-accelerated AI/ML and DL threats analyzed in the withdrawn 2024 Preprints paper, but also the potential use of AI to enhance both quantum algorithms and quantum computers. [8] Some of the different ways in which this synergistic relationship between AI and quantum-acceleration can impact different aspects of both quantum hardware and software are shown in Figure 3.

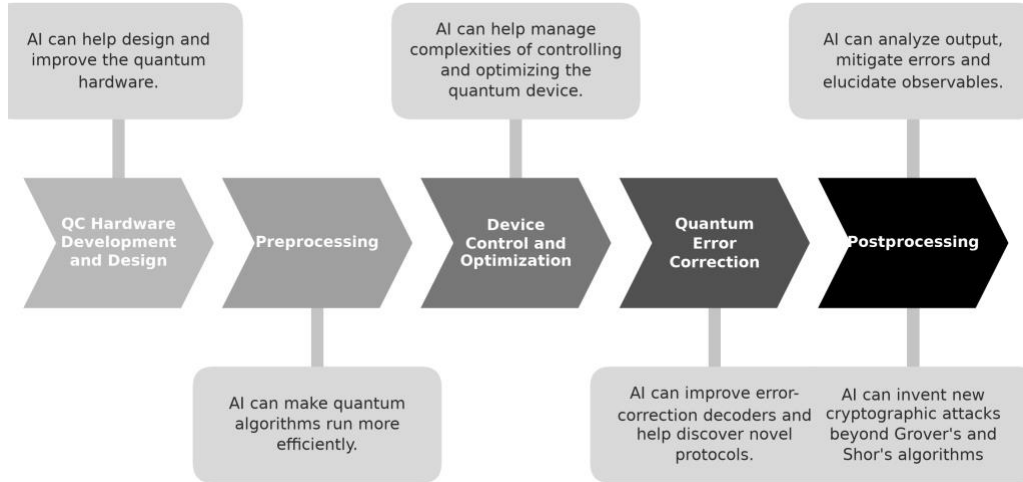


Figure 3: Bi-Directional Impact of Quantum and AI Exponential Development

This paper will collectively refer to such vulnerabilities as Cryptographically Relevant Quantum Accelerated AI and AI-Enhanced Quantum (CRQC+AI) threats. This paper investigates the most recent issues impacting the CRQC+AI vulnerability timelines of PQC adoption as part of the TLS 1.3 protocol and advocates for more research and development of crypto-agility for quantum resilient hybrid cryptographic frameworks capable of withstanding CRQC+AI attacks and gracefully migrating away from compromised ciphers. [9] [167] [168] [169] [173]

Starting with quantum algorithms, this paper explores cryptographically relevant algorithms that extend beyond the well-known Grover's algorithm [10] and Shor's algorithms. [11] At the forefront of these is the Harrow-Hassidim-Lloyd (HHL) algorithm, a quantum algorithm designed to efficiently solve certain systems of linear equations, particularly those involving sparse matrices with low condition numbers. [12] Initially perceived as theoretical, recent advancements in AI suggest that practical implementations of HHL could become reality sooner than previously expected. [13] The PQC encryption algorithms first approved by NIST use lattice-based cryptography. [14] This frequently involves solving linear equations and can be reliant on the Learning with Errors (LWE) problem, [15] which raises the question, examined and substantially qualified in Section 2.1, of whether AI-enhanced quantum linear-algebra methods like HHL could erode the margins of lattice-based cryptography and, with it, PQC as a defense against Harvest-Now, Decrypt-Later (HNDL) attacks. [16]

Other issues this paper investigates in Section 2.2 include a combined quantum-geometric approach to cryptanalysis that integrates CRQC+AI with low-dimensional representations in geometric algebra. Low-dimensional representations in geometric algebra simplify complex cryptographic problems into geometrically accessible forms, enabling CRQC+AI optimization techniques such as quantum annealing or

variational quantum eigensolvers as examined in Sections 2.3 and 2.4 to more effectively identify and exploit vulnerabilities within cryptographic schemes, including lattice-based PQC. Numerous additional recent CRQC+AI approaches, including side-channel attacks, are also surveyed in Section 3.

Many quantum hardware limitations continue to be resolved; as Section 2.1 details, however, some of the hurdles facing HHL (notably the density and conditioning of the matrices arising in LWE) are intrinsic to the problem rather than temporary engineering limits. The pairing of CRQC linear algebra methods such as HHL with the AI geometric simplifications provided by low-dimensional representations in geometric algebra is investigated here as a potential cryptanalytic attack vector against PQC. As Sections 2.1 and 2.2 establish, no mechanism for such a break is presently demonstrated. The HHL approach is considered as a polynomial acceleration effort, and the geometric algebra approach is treated as an exploratory, falsifiable hypothesis rather than an established threat. The paper concludes with an exploration of various AI-generated CRQC+AI vulnerability timelines for both RSA/ECC and lattice-based PQC encryption algorithms in Sections 3 and 4. These sections demonstrate that the CRQC+AI risk profiles are more severe and accelerate sooner than originally projected, underscoring the critical need for immediate action toward crypto-agile hybrid cryptography as a defense.

The contribution of this paper is therefore twofold: (i) an evidence-graded evaluation of the CRQC+AI attack surface against TLS 1.3 that separates mechanism-backed threats (Shor against RSA and ECC) from contingent ones (the lattice schemes, exposed only under a falsifiable structural collapse), and also from hypothesis-only ones (hash-based and symmetric primitives); and (ii) an AI-accelerated timeline model projecting runtime attack estimates and when a 50% likelihood crossover plausibly occurs for each of these attack surfaces against TLS 1.3 under different scenarios. Of these, the timeline model is the paper’s primary methodological contribution: a reproducible, parameterized scenario-estimation instrument in which the parameters can be varied and the model rerun, and whose explicit update mechanics (Sections 3.3.5 and 5) specify how future observations that follow, or deviate from, the modeled curves that enable potential revisions to specific parameters such that successive runs of the model track progressively closer to the accumulating historical record. In this regard, the paper contributes an instrument built to be recalibrated, rather than a one-time forecast, to the quantum resource-estimation work, [79] [82] expert-elicitation timeline studies such as the Global Risk Institute quantum threat reports, [80] [83] and quantitative risk-assessment methodology. The combined findings demonstrate that even the earliest and best grounded attack surfaces can arrive sooner than the 2030 to 2035 PQC migration window originally assumed. These findings should motivate crypto-agility as a PQC migration strategy, rather than implementation of single or even static-hybrid encryption algorithms as a PQC migration response. To support external scrutiny, the model is reproducible from explicit parameters and pseudocode (Appendix A), accompanied by a parameter sensitivity analysis (Appendix A.4), and paired with an explicit falsification section (Section 5).

Adopting hybrid cryptography and crypto-agile frameworks that can withstand CRQC+AI threats and gracefully migrate away from compromised ciphers is now more essential than ever. Equally critical is intensifying research into resilient cryptography solutions to mitigate against HNDL attacks and extend the CRQC+AI vulnerability timeline for TLS 1.3 well beyond the next decade. All of the authors have some affiliation with EnQuanta, which develops commercial quantum-resilient and crypto-agile cryptographic products; this interest is detailed in the Conflicts of Interest statement.

To keep the evidence base of this analysis explicit, the CRQC+AI threats analyzed in this paper fall into three tiers that are distinguished by how firmly each is grounded in a known mechanism (Table 1A).

Tier	Algorithms / primitives	Mechanism status	Paper’s correct claim
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Mechanism-backed	RSA, ECC	Shor’s algorithm gives a known polynomial-time quantum attack once sufficient fault-tolerant quantum resources exist	Migration away from RSA and ECC is mandatory
Contingency-backed	ML-KEM (Kyber), ML-DSA (CRYSTALS-Dilithium)	Requires an undiscovered reduction in effective lattice hardness, modeled here as the collapse parameter ρ	Conditional stress scenario, not a known break
Hypothesis-only	SLH-DSA, AES-256	No current structural attacks: Grover-only analysis is already accounted for	Included only as unknown-unknown risk, not a forecast

Table 1A: Evidence tiers for the quantum and CRQC+AI threats to TLS 1.3 cryptographic dependencies.

This paper does not claim a known break of current standardized PQC algorithms in TLS 1.3. It presents a scenario-based stress test of how much quantum hardware progress, AI-assisted engineering, and hypothetical cryptanalytic structure would be required for earlier-than-expected vulnerability compromise. The paper supports a conclusion that the rational response to the CRQC+AI vulnerability spectrum for TLS 1.3 is to encourage crypto-agile replaceable deployment of PQC in a hybrid cryptographic framework.

Tier	Confidence a reader may place	Promotes when	Demotes when
Mechanism-backed (RSA, ECC via Shor)	High; a standard result	(not applicable)	(not applicable)
Contingency-backed (lattice under ρ -collapse)	Conditional; holds only if the collapse is discovered	a geometric embedding with $\rho > 0$ is demonstrated	the conjecture is refuted, or a lower bound closes it
Hypothesis-only (SLH-DSA, AES-256)	None as a dated claim	a concrete attack is published	remains undated

Table 1B: Assurance and promotion/demotion by evidence tier (complements Table 1A).

The promotion and demotion triggers above follow the evidence rule of Section 3.3.5; they should be read as the conditions under which a reader may raise or lower the confidence attached to each tier, not as predictions that those conditions will be met.

1.1 Scope and Exclusions

In scope. This paper’s instrument estimates when five cryptographic primitives, RSA-2048, ML-KEM768/1024, ML-DSA-44, SLH-DSA and AES-256, as deployed in TLS 1.3, become feasibly attackable under combined quantum and AI acceleration. It measures primitive erosion.

Out of scope. The instrument does not model information technology (IT) organizational estate architecture, inventory, rotation authority or change control, nor does it transfer to other protocols. TLS 1.3 is deliberately the most tractable migration surface in the IT estate: keys are ephemeral and rotatable, algorithms are negotiated, and certificates are short-lived. SSH, IPsec, DNSSEC, and code and firmware signing carry materially worse migration profiles (long-lived or unrotatable keys, trust anchors with decade-plus lifetimes, and size constraints), and the paper’s Cloudflare citation, showing over half of human-initiated traffic already post-quantum encrypted, indicates how far ahead TLS already is.

Conclusions about primitive erosion in TLS 1.3 therefore represent a best case and do not transfer to those surfaces, or to estate-level recommendations, without the additional analysis in the sections below.

1.2 Approach and Instrument Map

Approach. Three instruments are used, in this order. First, the channel model (Section 3.3) decomposes engineering progress toward a CRQC into physical-qubit count, gate fidelity, code efficiency and decoder latency, and yields the hardware budget $B(t)$. Second, the capability model (Appendix A) generates the runtime and feasibility curves from two exponential drivers, one for hardware and one for AGI-gated software. Third, the cost model (Section 4.5) restates the lattice and hash contingencies in explicit cryptanalytic units.

Instrument map. The capability model generates a cost model and channel model check. The cost model's collapse parameter ρ and the capability model's software couplings are two parameterizations of one conjecture, an undiscovered effective-dimension collapse, so agreement among them is consistency, not independent corroboration (see Section 3.3.4). What was done, in what order, on what evidence, and what was excluded is stated in the Scope section above.

2. Background

This section reviews the CRQC+AI techniques that bear on the attack pathways analyzed later in the paper. It includes methods such as quantum annealing and the variational quantum eigen-solver not because they break standardized cryptography on their own, the analysis below concludes they do not; but because establishing what does not contribute to a near-term break is as important to the argument as identifying what might.

2.1 Harrow-Hassidim-Lloyd Background

The Harrow-Hassidim-Lloyd (HHL) algorithm is a quantum algorithm designed to solve systems of linear equations exponentially faster than classical methods under certain conditions. This efficiency arises when dealing with sparse matrices that have low condition numbers, meaning the ratio of the largest to smallest eigenvalues is small.

Lattice-based encryption schemes, such as those based on the LWE problem, rely on the computational hardness of certain lattice problems, which can often be reduced to solving linear equations. The rapid advancement of quantum algorithms such as HHL and the linear combination of Hamiltonian simulation (LCHS) technique introduces a growing area of concern for these cryptographic systems. [17] The most prominent recent attempt in this direction, a 2024 claim of a polynomial-time quantum algorithm for LWE, was withdrawn by its own author within days after a bug was identified, which underscores that no quantum attack has yet overturned lattice hardness. [18] If quantum hardware evolves to efficiently implement the HHL algorithm, it could dramatically accelerate the solution of linear equations underlying lattice problems, posing a serious threat to the security of lattice-based encryption.

Quantum hardware barriers such as error rates and limited scalability are easing steadily, with advances in error correction and hybrid quantum-classical techniques narrowing the gap between theory and practice. [19] [20] Not every obstacle facing HHL is of this kind, however. The requirement for sparse, well-conditioned matrices is a property of the problem instance rather than of the hardware, and as the formal analysis below shows, it is precisely where lattice problems fail to cooperate.

Formally, for an $N \times N$ linear system $Ax = b$ with sparsity s and condition number κ , solved to precision ε , the HHL algorithm and its near-optimal refinement by Childs, Kothari, and Somma [21] run in time:

$$T_{\text{HHL}} = \tilde{O}(s \cdot \kappa \cdot \text{polylog}(N/\varepsilon)) \text{ [near-optimal in } \varepsilon]$$

$$T_{\text{HHL}} = \tilde{O}((\log N) \cdot s^2 \cdot \kappa^2 / \varepsilon) \text{ [original HHL]}$$

The decisive structural features are that the runtime is polylogarithmic in the dimension N but polynomial in the sparsity s and the condition number κ , and that the output is a quantum state $|x\rangle$ proportional to the solution rather than the classical solution vector. Useful information is obtained by estimating an observable $\langle x|M|x\rangle$; recovering the individual solution coordinates removes the exponential advantage. For LWE the relevant system matrices are effectively dense (s on the order of N) and not well-conditioned, and a key-recovery attack must read the secret out coordinate by coordinate. In addition, LWE is not an exact linear solve but a bounded-distance decoding problem (recovering the secret from $As + e = b$ with small noise e), so the clean $Ax = b$ formulation that HHL assumes does not directly apply. An unmodified HHL therefore does not by itself break LWE; its role, together with the LCHS technique for non-unitary dynamics, is confined to accelerating polynomial factors of an inner linear-algebra step. The cost model of Section 4.5 makes this boundary explicit. As with the annealing and variational methods discussed below, HHL is treated here as a candidate attack capability to be evaluated on its merits rather than assumed. The finding that it contributes at most polynomial acceleration to an inner step, rather than a standalone break, is reported as deliberately as any positive result and forms part of the due-diligence basis for the timeline analysis that follows.

2.2 Low-Dimensional Geometric Algebra Background

The intersection of CRQC+AI and low-dimensional representations in geometric algebra presents intriguing theoretical pathways to breaking both current cryptographic systems and PQC encryption. [22] Unlike the methods in Sections 2.1, 2.3, and 2.4, which rest on established algorithms with known performance bounds, the geometric-algebra approach is presented as an exploratory hypothesis rather than a demonstrated capability: no mechanism is presently known by which it yields a concrete cryptanalytic advantage. It is included, and evaluated, as a candidate direction precisely so that its speculative status is explicit and falsifiable rather than assumed, on the same due-diligence basis applied to the other methods in this section.

2.2.1 Quantum AI-Driven Optimization of Cryptanalysis

The first idea is to apply quantum optimizers such as quantum annealing and the variational quantum eigensolver, treated in Sections 2.3 and 2.4 below, to a geometric encoding of a cryptographic problem, in the hope that a low-dimensional geometric form exposes structure that an unstructured search would miss. The limitation carries over from those sections: such optimizers carry no demonstrated scaling advantage on unstructured problems, so this step helps only if the geometric encoding itself supplies exploitable structure, which is the open question rather than a settled result.

2.2.2 Geometric Algebraic Representation of Cryptographic Spaces

The load-bearing idea is representational. If a lattice or its underlying linear system could be embedded in a low-dimensional geometric algebra, operations in that space might expose linear dependencies or simplifications that are invisible in the standard basis. [23] Everything else in this section depends on

whether such an embedding exists and is efficiently computable, and no construction achieving this against standardized lattices is presently known.

2.2.3 Breaking PQC with Quantum-Geometric Hybrids

Our precise conjecture is that such an embedding would shrink the lattice’s effective block size β , the parameter that sets the cost of sieving, an effect captured by the collapse fraction ρ defined at the close of this section and quantified in Section 4.5. This is the single quantity on which any geometric quantum break of lattice PQC would rest; this paper states how large ρ would need to be rather than asserting that value is attainable.

2.2.4 Geometrically Inspired Quantum AI Algorithms

A more speculative variant uses quantum machine learning over the geometric representation to learn structural weaknesses from data rather than derive them analytically. It shares the trainability and classical simulation limits discussed for the variational methods in Section 2.4 and has no demonstrated application to standardized PQC. It is noted for completeness, not advanced as a near-term threat.

2.2.5 Parameterized Geometry as an Attack Vector

Finally, large AI models could in principle search systematically over geometric representations and their parameters, an automated hunt for an embedding that yields a nonzero ρ . This describes how such an attack might be discovered, not evidence that it exists. Absent a known mechanism, the parameterized geometry search has no established target.

In summary, the geometric-algebra direction is a single hypothesis presented in several forms: namely, that some low-dimensional embedding could expose structure in a lattice that classical algebra does not. Its entire cryptanalytic force reduces to whether such an embedding can shrink the effective hardness of the lattice, and no construction achieving that against standardized schemes is currently known. The section is retained not as evidence of a break, but rather as an explicit, falsifiable hypothesis, stated precisely enough in the paragraph that follows to be tested or discarded.

To make this hypothesis precise enough to test, this paper represents the conjectured effect of a low-dimensional geometric embedding as a reduction of the operative lattice block size, written $\beta \rightarrow \beta(1 - \rho)$ for a collapse fraction $\rho \in [0, 1)$. No mechanism is presently known that achieves $\rho > 0$ against the random q -ary lattices underlying standardized LWE; ρ is introduced here as an explicit, falsifiable parameter rather than an established result. Section 4.5 quantifies the value of ρ that the modeled 2032 to 2035 lattice crossover vulnerabilities would require.

2.3 Quantum Annealing Background

By way of framing, quantum annealing is presented here as background: the analysis in Sections 3.3.5 and 4.5 concludes that it does not, on its own, yield a standalone cryptanalytic break of the schemes considered in this paper. It is surveyed as part of the adjacent quantum-optimization landscape, alongside the variational methods of Section 2.4, rather than as a demonstrated attack path.

Quantum annealing is a specialized quantum computing technique used to solve complex optimization and combinatorial problems by leveraging quantum superposition, entanglement, and tunneling. [24] It is based on the principles of adiabatic quantum computation (AQC). [25] A quantum system implementing

AQC is initialized in the ground state of an easy-to-prepare Hamiltonian [26] and then slowly evolved into a Hamiltonian that encodes the solution to a given optimization problem.

2.3.1 Key Aspects of Quantum Annealing

2.3.1.1 Energy Minimization Approach Quantum annealing works by finding the lowest energy state (global minimum) of an objective function mapped onto a quantum system, commonly represented as a mathematical optimization problem.

Concretely, the objective is encoded in a problem Hamiltonian H_P whose ground state is the optimal solution. In Ising form,

$$H_P = \sum_i h_i \sigma_{iz} + \sum_{i < j} J_{ij} \sigma_{iz} \sigma_{jz}$$

or equivalently as a quadratic unconstrained binary optimization (QUBO) problem, the minimization over $x \in \{0,1\}^n$ of $x^T Q x$. Cryptanalytic search problems are attacked by mapping them into this form.

2.3.1.2 Quantum Tunneling Unlike classical annealing methods (such as simulated annealing) that rely on thermal fluctuations to escape local minima, quantum annealing can leverage quantum tunneling to explore multiple solutions simultaneously, potentially leading to a faster convergence.

The advantage is sharpest for tall, narrow barriers: the quantum tunneling rate through a barrier of height h and width w scales as $\exp(-w h)$, whereas the thermal hopping used by classical simulated annealing scales as $\exp(-h / k_B T)$. Where the energy landscape has thin barriers, tunneling can traverse them at a rate the thermal process cannot match.

2.3.1.3 Hamiltonian Evolution The system starts with a simple Hamiltonian whose ground state is known. Over time, this Hamiltonian is gradually transformed into a more complex problem-specific Hamiltonian while ideally maintaining the ground state.

Formally the system follows a time-dependent interpolation between an easy initial Hamiltonian H_0 (with $H_0 = -\sum_i \sigma_i^x$, a transverse field whose ground state is an equal superposition) and the problem Hamiltonian H_P :

$$H(s) = (1 - s) \cdot H_0 + s \cdot H_P, s = t/T, t \in [0, T]$$

By the adiabatic theorem the system stays in the ground state only if the evolution is slow relative to the inverse square of the minimum spectral gap:

$$T \gtrsim \max_{0 \leq s \leq 1} |\langle E_1(s) | dH/ds | E_0(s) \rangle| / g_{\min}^2, g_{\min} = \min_s [E_1(s) - E_0(s)]$$

The runtime is therefore governed by $g_{\min}$. For benign landscapes the gap stays open and annealing is fast, but for the hardest combinatorial and cryptanalytic instances the gap can close exponentially in the problem size n , in which case the required annealing time grows exponentially, and no quantum speedup is obtained. This gap-dependence is the precise reason quantum annealing offers advantage on some structured problems while providing no general guarantee against well-designed cryptography.

2.3.1.4 Quantum Superposition & Entanglement The qubits in a quantum annealer exist in a superposition of states and can be entangled, allowing for parallel exploration of multiple solutions.

2.3.2 Specialized Hardware

Companies like D-Wave Systems have developed quantum annealers that use superconducting qubits for optimization problems. [27] Unlike universal quantum computers, which perform arbitrary quantum algorithms, quantum annealers are primarily tailored for solving specific optimization and machine learning tasks.

2.3.3 Applications of Quantum Annealing

Quantum annealing is applied wherever a problem can be cast as energy minimization over a discrete configuration space, typically by encoding it as a quadratic unconstrained binary optimization (QUBO) or Ising instance and letting the annealer relax toward a low-energy state. [28] The three domains below are the most developed. Practical, if still modest, value has appeared in the first two; the third is the domain most relevant to this paper and, at present, also the most speculative. We treat annealing, like the variational eigensolver methods in Section 2.4, as a candidate attack capability to be evaluated on its merits rather than assumed. We report the negative findings as deliberately as the positive ones, because ruling a method out is itself part of the due-diligence basis for the timeline analysis that follows.

2.3.3.1 Optimization Problems Logistics, scheduling, portfolio optimization, traffic flow management.

These are the workloads quantum annealing was built for, and they share a common shape: a combinatorial objective with many competing constraints that maps naturally onto a QUBO or Ising cost function. Vehicle routing and traffic flow have been the most visible pilots, including Volkswagen’s traffic-routing demonstrations, [29] while portfolio optimization recasts the Markowitz mean-variance objective as a binary selection problem, [30] and scheduling problems (job-shop, crew, and gate assignment) translate their constraints into penalty terms in the Hamiltonian. [31] The practical caveat is that, although there is a clear advantage over mature classical heuristics in these pilots, such as simulated annealing, tabu search, and modern mixed-integer solvers, any advantage has not been established in general. Problems where quantum annealing helps tend to be on landscapes with tall, narrow energy barriers that favor tunneling over thermal hopping, and the benefit is instance-specific rather than universal. Embedding the logical problem onto the fixed hardware graph (e.g., the minor-embedding step on the Chimera and Pegasus topologies) also consumes physical qubits rapidly, which currently bounds the size of problems that can be run natively.

2.3.3.2 Machine Learning feature selection, clustering, and deep learning acceleration.

In machine learning the annealer is used in two ways: as an optimizer and as a sampler. As an optimizer it handles discrete subproblems such as feature selection, balanced clustering, and the training of structured models, each expressed as a QUBO. [32] These are proof-of-concept demonstrations rather than evidence of advantage; the early deep-network training results, for example, were small and coarse-grained and were subsequently matched by classical methods. As a sampler it draws low-energy configurations that approximate a Boltzmann distribution, which has been used to train restricted and quantum Boltzmann machines and to seed generative models. [33] This sampling role carries an important qualification: the hardware does not produce a clean quantum Boltzmann distribution but something closer to a classical distribution at an unknown, instance-dependent effective temperature, distorted by freeze-out during the anneal, and the quantum Boltzmann machine itself remains largely a theoretical and simulation construction rather than a demonstrated hardware advantage. The appeal is that sampling from complex distributions is expensive classically, so a hardware sampler could in principle accelerate part of a learning pipeline. The same quantum hardware limits apply, however: restricted qubit counts, sparse connectivity,

embedding overhead, and analog control noise keep demonstrations small. On standard benchmarks, classical machine learning remains dominant over currently known annealer implementations. For the threat model of this paper the relevance is indirect: annealing-assisted learning could speed components of a cryptanalytic pipeline, for example extracting structure from side-channel traces, but it acts as a contributing tool rather than a standalone capability.

2.3.3.3 Cryptography & Cybersecurity Enhancing cryptographic algorithms and security models.

On the offensive side, cryptanalysis can be recast as optimization and handed to an annealer. Integer factorization is the canonical example: writing $N = p \cdot q$ and minimizing $(N - p \cdot q)^2$ over the binary digits of the factors turns factoring into a QUBO, and small semiprimes (numbers ranging from the hundreds up to the hundreds of thousands) have been factored this way on annealing hardware, usually with substantial classical preprocessing. [34] Lattice problems have received similar treatment, with the shortest-vector and closest-vector problems, and by extension LWE, formulated as Ising or QUBO instances for both annealers, [35] as well as quantum enumeration variational solvers. [36] These are existence results rather than evidence of scaling; the gate-model lattice work, for instance, solves only a minority of instances even in small dimensions. In every case the same two limits recur: the qubit count and connectivity required grow rapidly with problem size. There is no scaling advantage over the best classical attacks (the general number field sieve for factoring, lattice reduction for LWE) that has been demonstrated at or near cryptographic parameters. Current analyses of RSA-scale factoring on a D-Wave annealer document the wall rather than a breach. [37] This is consistent with the gap-dependence established in Section 2.3.1.3: because the minimum spectral gap can close exponentially for hard, well-structured instances, annealing carries no general guarantee against properly parameterized cryptography. On the defensive side, the same optimization machinery is applied constructively, to network-security resource and sensor placement, intrusion-detection feature selection, anomaly detection, and attack-graph and vulnerability analysis. The honest summary, and the one this paper adopts, is that quantum annealing is a component of the broader CRQC+AI optimization toolkit rather than a standalone route to breaking standardized cryptography.

In summary, while quantum annealing is promising for solving certain classes of problems, it does not provide exponential speedups for all computational tasks. However, it has shown potential advantages in real-world optimization and AI-related challenges in areas like cryptography and cybersecurity.

2.4 Variational Quantum Eigensolver Background

The Variational Quantum Eigensolver (VQE) is a hybrid quantum-classical algorithm designed to find the ground state energy of a quantum system, particularly useful for solving problems in quantum chemistry and materials science. [38] It combines quantum computing with classical optimization to efficiently approximate the lowest eigenvalue (ground state energy) of a given Hamiltonian.

2.4.1 How VQE Works

VQE parameterizes a quantum circuit, the ansatz $U(\theta)$, and optimizes it with a classical loop. The target Hamiltonian is written as a weighted sum of Pauli operators, $H = \sum_k c_k P_k$, a form the quantum device can measure directly. The ansatz prepares a parameterized state $|\psi(\theta)\rangle = U(\theta)|0\rangle$, and the variational principle guarantees that its energy expectation $E(\theta) = \langle\psi(\theta)|H|\psi(\theta)\rangle$ is bounded below by the true ground-state energy E_0 , with equality when the ansatz reaches the ground state. The device assembles $\langle H \rangle$ from the

separately measured Pauli terms, a classical optimizer minimizes $E(\theta)$ over θ , obtaining gradients directly on hardware through the parameter-shift rule, and the loop repeats until the energy converges.

Two limitations bound this approach in the cryptanalytic setting. The optimization landscape is subject to barren plateaus: for many ansatz families the gradient $\partial E/\partial \theta$ vanishes exponentially in the qubit count, so the variational search becomes effectively untrainable at the scale a cryptographic problem would demand. [39] More fundamentally, VQE is a ground-state estimation method for physical and chemical Hamiltonians and provides no established reduction from integer factoring, discrete logarithm, or lattice problems. It is surveyed here as part of the quantum-optimization landscape, not as a demonstrated route to breaking standardized cryptography.

2.4.2 Key Advantages of VQE

VQE's appeal on near-term hardware rests on three properties [40]. First, it is hybrid-efficient: the quantum processor runs only short, shallow circuits to prepare the trial state and measure Pauli expectations, while the classical optimizer carries the search, so the hardware does only what is classically hard. [38] [41] Second, it is comparatively noise-resilient, because the variational principle tolerates the small state-preparation and readout errors that would corrupt quantum phase estimation, and dedicated error-mitigation techniques can extend its usable reach on noisy devices [42] [43]. Third, the ansatz is a free design choice, ranging from hardware-efficient circuits matched to a device's native gates, [44] to adaptive, problem-tailored constructions that grow the circuit only as needed. [45] These same properties bound its cryptanalytic power: the shallow circuits and heuristic optimization that make VQE practical, together with the trainability limits noted above [46], are also what keep it from performing the deep, structured computation that an attack on factoring, discrete-logarithm, or lattice problems would require.

In summary, VQE is one of the most promising algorithms for Noisy Intermediate-Scale Quantum (NISQ) devices, offering a practical approach to leveraging quantum computation before fully error-corrected quantum computers become viable. [47]

3. CRQC+AI Vulnerability Timelines

3.1 Current Timeline Projections

The discussion of timelines in this section should be understood as the authors' best current estimates based on the factors discussed. In presenting these timelines, we acknowledge our inherent limitations to accurately estimate timelines of future technologies. Humans have always been challenged in making predictions as our abilities to predict are rooted in linear projections, not the exponential growth of accelerating change. [48]

Current timeline projections for when RSA, ECC, and similar factoring- and/or elliptic-curve-based public key encryption algorithms need to be replaced by lattice-based PQC encryption algorithms have focused primarily on the growth of quantum computing processing power evaluated in terms of numbers of logical gates in conjunction with the use of Shor's and Grover's quantum algorithms. [49] The approach used by NIST with respect to timelines for when new generations of encryption solutions should be adopted in the face of threats from quantum computers is based on the principles of Mosca's Theorem. [50] This approach correlates the intended duration of data protection and the length of time needed for an organization to implement a new encryption solution in view of the potential for HNDL attacks. [51]

Mosca expresses this balance as a simple inequality. Let X be the length of time the data must remain confidential, Y the time an organization needs to migrate to new cryptography, and Z the time until a cryptographically relevant quantum computer exists. Then whenever

$$X + Y > Z$$

the organization is already exposed, because data harvested today under HNDL attacks may still be sensitive by the time the capability to decrypt it arrives. The practical force of the inequality is that it does not require a known value of Z : any statistically significant probability of a break within the secrecy horizon X is sufficient to make migration and crypto-agility rational today. NIST's general expectation, though never stated as a formal prediction, was that PQC encryption would serve as a replacement for RSA and similar algorithms for multiple decades after RSA encryption is compromised. [52] Unfortunately, even as an informal expectation this durability now appears more wishful than analytical. Importantly, none of these expectations, nor the formal projections for RSA encryption itself, accounted for the accelerating changes of CRQC+AI seen in the last few years and the potential for AGI in the next few years. [53] [54] [55] [56] [57]

The current approach to timeline projections for these CRQC+AI vulnerabilities is sometimes referred to as Q-Day, which analogizes to Y2K and assumes the single-point deadline based on a single-point failure. [58] Google recently bumped its PQC transition time frame goals from 2030 to 2029. [59] As noted by IBM in a recent article, however, the better approach is not to assume Q-Day is a single day for a single event, but rather to model a spectrum of risks that will have a range of dates for ranges of risks across a range of breaches across a spectrum of different types of cryptographic algorithms. [60] More recent IBM articles characterizes the challenge presented by the PQC transition as the most significant computing revolution in 60 years and as the digital equivalent of needing to change the locks and keys on every single door in the world. [61] [173]

Accordingly, the CRQC+AI spectrum projections below are model-generated scenario estimates, not empirical measurements. The projections are based on a range of factors that form a CRQC+AI risk band plotted over time between a lower bound based on linear extrapolation of potential CRQC improvement based on known quantum and AI developments and an upper bound based on an exponential extrapolation of potential CRQC+AI improvements based on an assumption that there is a meaningful probability that AGI can be achieved by the end of 2028. [62]

3.1.1 Historical Analogues for the PQC Transition: the 2008 Financial Crisis and Y2K

The appropriate historical analogue for a CRQC+AI cryptographic break is not Y2K alone. Y2K was a contained engineering problem: a single, calendar-fixed deadline whose cost was remediation, largely completed before the date passed. A cryptographic break has a different character. It is systemic, cross-institutional and confidence-driven: certificate trust, key establishment and signed software underpin essentially every digital relationship at once, and traffic harvested today can be decrypted retroactively long after the fact. The closer analogue for this kind of systemic failure mode is the 2008 financial crisis, in which the damage was dominated not by the direct cost of the failing instruments but by contagion, the collapse of counterparty confidence, and cross-institutional propagation. The systemic estimate this paper itself adopts, the Citi and Hudson Institute figure for a single-day attack on one major bank's payment-rail access, is of that order rather than of Y2K's remediation history. This section therefore leads with the 2008 analogue for the scale and character of the exposure and retains the Y2K comparison for the narrower and still-useful lesson it teaches, namely how a fixed, unpostponable deadline concentrates remediation, a property the PQC transition largely lacks.

Systemic and economic stakes. The scale implied by this analogue is not hypothetical. Independent modeling by the Hudson Institute (Butler, Prosperity at Risk, April 2023) [172], adopted in the Citi Institute's 2026 analysis [70], estimates the direct cost of a single-day quantum-enabled attack on one top-five U.S. bank's Fedwire access at roughly \$730 billion to \$1.95 trillion, and the indirect impact, the contagion cost measured as GDP-at-risk over a resulting recession, at \$2.0 to \$3.3 trillion, about 10 to 17 percent of GDP. The gap between the two is the point: as in 2008, the dominant cost is not the direct failure but the systemic contagion that follows a loss of confidence in the financial plumbing. This is a single modeled scenario on a single payment rail, offered as an order-of-magnitude indication rather than a comprehensive economic estimate; the sector-by-sector variation in exposure is taken up in Section 6.2.

Unlike the Y2K threat at the end of 1999 that was a single weakness based on a single critical date, the CRQC+AI threats underlying the PQC transition represent a broad spectrum of threats and risks. Even though these risks are orders of magnitude greater than risks tolerated for other kinds of high-impact fault events like nuclear power plant meltdowns, many of those who will be most impacted by these risks have been slow to realize, plan for, and implement the recommended strategies NIST has promulgated over the last decade. [63] The contrast in tolerated risk is stark: modern nuclear reactor designs are held to a core damage frequency approaching one in ten million per reactor year, [64] while the chief executive of a leading AI laboratory has publicly put the probability that advanced AI development goes catastrophically wrong at roughly one in four. [65] The two different kinds of threat risks are not measured on the same basis, yet the tolerated risk gap between a one in ten million engineering target for nuclear risks and a one in four expert estimate for catastrophic AI outcomes underscores how unevenly these risks are treated. This also explains why awareness and response for cyber risks follow an exponential pattern, with slow leadups to a dramatically increasing push as a deadline looms, or, even worse, after a crisis occurs.

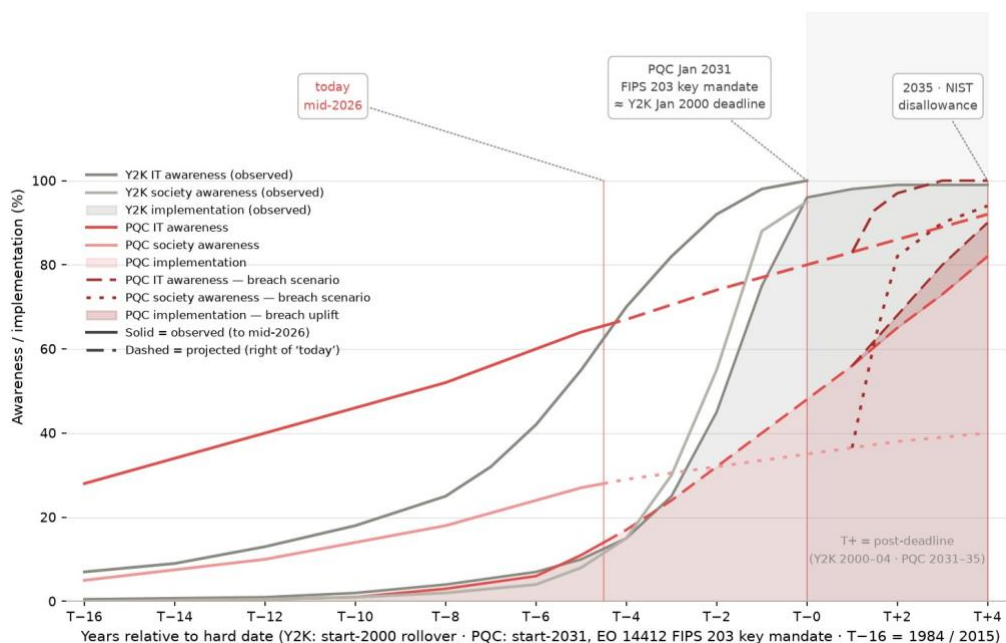


Figure 4A: Comparison of Awareness and Response for Y2K vs. PQC Transition

Figure 4A compares awareness and response for Y2K versus PQC transition on the relative scale of corresponding 20-year timelines. The starting dates at T-16 correspond to the equivalent historical origins for Y2K (1984) [66] and PQC (2015). [67] The similarity in both the public awareness and implementation

curves as of today (mid-2026 at T-4.5) provides historical credibility for this comparison, even though both PQC awareness curves have led their Y2K equivalents up to this point. In comparing these two threat response timelines, it is important to note that the Y2K portions are entirely historical, whereas the PQC portions after today (mid-2026) were generated by the Anthropic Claude Fable 5 AI model in July 2026 based on various government reports and expert projections. [5] [51] [68] [69] [70] [71] [72] [73] The figure is rendered from a deterministic generation script so that the curves can be regenerated and audited; curve values are illustrative reconstructions calibrated to documented milestones and point-in-time survey data, not measured time series. The Y2K awareness curves are truncated at T-0 for legibility. It should also be noted that public attention to Y2K threats collapsed to roughly 20% after the uneventful rollover. It is also interesting to note that the PQC projections without a publicly disclosed breach scenario match our human inclination toward linear projections, whereas historical Y2K responses and the publicly disclosed breach scenario (est. in 2032) match the technological curve of increasing exponential growth. Although the two threats share a similar 20-year arc that spans early insider warnings, mid-course disclosure and standardization of responses, and endgame sprints, the PQC transition differs from Y2K in four decisive respects.

3.1.1.1 Deadlines. The Y2K deadline of T-0 was an entirely calendar-driven deadline (Jan. 1, 2000) that could not be postponed, which drove remediation to effective completion. The PQC 2031 transition deadline is entirely regulatory and is chosen by fiat as a single date (Jan. 1, 2031), which is the day after the effective date of the Dec. 31, 2030, EO 14412 mandate for FIPS 203 (ML-KEM) key establishment. [5] This single date is chosen as the best reasonable approximation of the single Y2K date across what is a broad spectrum of both regulatory deadlines from different governmental organizations and risk profile dates for RSA-2048 as presented in Section 4. The harder PQC cutoff is 2035, when NIST disallows classical RSA/ECC algorithms, which puts the estimated publicly disclosed breach scenario (est. in 2032) inside and in the middle of the expert-estimated CRQC+AI threat emergence window of 2029 to 2035.

3.1.1.2 Awareness. Y2K produced genuine mass concern (>85% of the public in 1999), which created bottom-up pressure on institutions. [74] PQC may not attract a similar mass concern. [75] Public recognition of the PQC cryptographic threat is approximately 25 to 30% today, with the threat effectively invisible to roughly 3 out of 4 ordinary people [75]. This means that pressure for remediation must be driven primarily by regulation and will not be driven by public recognition until after there is an actual publicly disclosed breach.

3.1.1.3 Breach. The breach scenario for RSA-2048 is shown in Figure 4A as a reasonable example with two important caveats: (i) it is an estimated projection, in this case based on an approximately 75% modeled feasibility estimate from Figure 6B at a projected attack runtime of between a day and a month per Figure 6A; and (ii) as noted in the Citi report [70] it assumes that a breach of RSA-2048 would become publicly disclosed as of or shortly after the breach occurred, which is an unlikely assumption if the breach was the result of actions by a state-actor. It must also be noted that even a Y2K-1999-scale emergency response to a publicly disclosed breach does not reach 100% PQC implementation by 2035. Unlike Y2K, the scope and pervasiveness of systems requiring PQC upgrades (protocols, PKI roots, HSMs, embedded devices, and long-lived OT hardware) make it unlikely without significant new innovations that all these systems could be breach-migrated in just three years. In addition, the uplift in implementation assumes that it is publicly disclosed. A covert-breach variant from the Citi report would leave all three dark-red curves collapsed onto their baselines, which is the report's most severe case. The stakes are material: the Citi Institute analysis,

drawing on Hudson Institute modeling from 2023, places the GDP at risk from a singleday quantum-enabled attack on the Fedwire access of one of the top five U.S. banks at \$2.0 to \$3.3 trillion. [70]

3.1.1.4 Cost. The cost of remediation efforts for Y2K was in the hundreds of billions-dollar scale: ~\$100B in the U.S. alone and an estimated \$300B to \$500B+ globally. [74] [76] Direct spending forecasts for the commercial PQC market (software, hardware, and services) are substantially smaller than the likely total migration cost. Future Market Insights projects the PQC migration market to grow from approximately \$1.9B in 2025 to \$12.4B by 2035. [77] Large-scale migration, however, requires inventory discovery, software redevelopment, certificate replacement, testing, compliance, hardware refresh, and operational deployment across virtually every digitally connected enterprise: Moody's estimates that PQC migration may consume roughly 2.5% of enterprise IT budgets, [78] while Citi characterizes the transition as a trillion-dollar security race for its systemic economic implications rather than as a quantified migration budget. [70] These figures measure different quantities (market size, migration spending, and economic exposure) and should not be read as a single global cost of replacing vulnerable cryptography. The response to Y2K is unfortunately not a reassuring precedent for our ability to respond to this kind of cyber-threat level event; but rather this comparison is a warning about the scale, time, and effort of what a full-scale remediation effort for the PQC transition will most likely require from a future historical perspective.

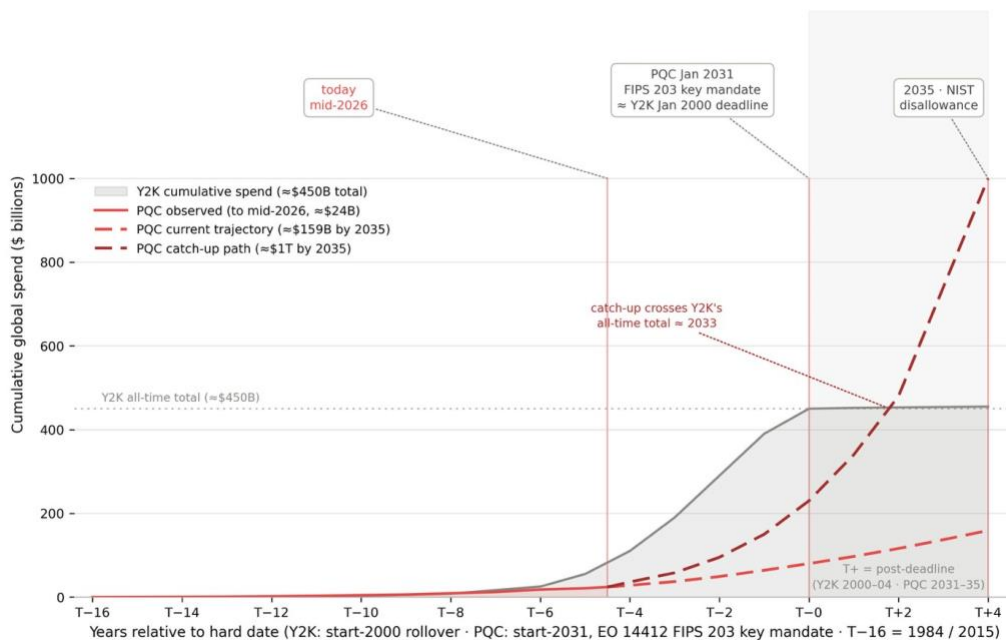


Figure 4B: Comparison of Cumulative Global Spend for Y2K vs. PQC Transition

3.1.2 RSA/ECC Vulnerability Spectrum

An academic analysis first posted in 2019 and published in 2021 estimated that 2048-bit RSA encryption could be broken in about 8 hours using Shor's algorithm and a fault-tolerant quantum computer built from roughly 20 million noisy physical qubits with surface code error correction. [79] In 2022, the White House directed NIST and the National Security Agency (NSA) to prioritize "the goal of mitigating as much of the quantum risk as is feasible by 2035." [63] Based on surveys of expert opinions of the growth in quantum

computing processing power into the 2030s, as of 2023 it was projected that the so-called Q-Day for RSA/ECC encryption was most likely to happen around the mid-to-late 2030s. [80]

But less than six months after NIST approved the first PQC algorithms in 2024, it became necessary to advance the projected timeline for deprecating RSA/ECC encryption for military network computers by five years. [5] At least one organization has an online countdown timer to the so-called Y2Q date for RSA/ECC that currently predicts this date to be less than four (4) years away. [81]

In May 2025, the same researcher at Google who predicted it would take 20 million qubits to break RSA-2048 in 2021 presented an approach for achieving a 20X reduction in this estimate. [82] Most recently in March 2026, the same research team at Google published a further reduction in their estimates [20], and in the same week moved its internal company deadline for PQC transition from 2030 to 2029. [59] In July 2026, an independent research team was purportedly able to exceed Google's results using a zero-knowledge proof approach. [170]

Two scaling relationships explain both why the estimates for potentially breaking these algorithms continue to fall and why the qubit estimates quoted below differ so widely. Shor's algorithm factors an n -bit modulus ($n = \log_2 N$) in a polynomial number of gates, on the order of $\tilde{O}(n^2)$ to $\tilde{O}(n^3)$, using roughly $2n + O(1)$ logical qubits, whereas the best classical method, the general number field sieve, runs in sub-exponential time:

$$L_N = \exp[(1.92 + o(1)) \cdot (\ln N)^{1/3} \cdot (\ln \ln N)^{2/3}]$$

It is this collapse from sub-exponential to polynomial that dooms RSA and ECC. The second relationship governs the gap between logical and physical qubits. Under the surface code each logical qubit consumes on the order of $2d^2$ physical qubits, where the code distance d is fixed by the target logical error rate:

$$p_L \approx A \cdot (p_{\text{phys}} / p_{\text{th}})^{(d+1)/2}$$

The widely differing figures cited below are therefore not contradictory: counts near 100,000 are physical-qubit estimates inclusive of error-correction overhead, while counts in the low thousands (down to 1,193 for ECC-256) are logical or idealized-circuit estimates taken before that overhead is applied. Comparing them requires fixing whether the code distance d , and hence the physical multiplier $2d^2$, has been included.

The compression of the various deadlines for PQC transition is reflected in the ever-increasing power of CRQC developments. The most recent briefing by Global Risk Institute reflects the implications of the continual reductions in estimated number of qubits needed for CRQC relevant to different NIST algorithms. [83] This report includes a figure whose caption reads: "Figure 1 (Cain et al.) [19] Estimated number of physical qubits needed to run Shor's algorithm compared over the years of publication for previous resource estimates and for Cain et al. Please note that the vertical scale is logarithmic: published resource estimates have dropped by orders of magnitude over time." Figure 5 below is based on these various published estimates of physical qubits required for Shor's algorithm by year of publication, with a fitted exponential decay dashed line representing the trendline. Data from Cain et al. [19] as compiled in the Global Risk Institute briefing [83]; figure redrawn by the authors, with the exponential decay fit added.

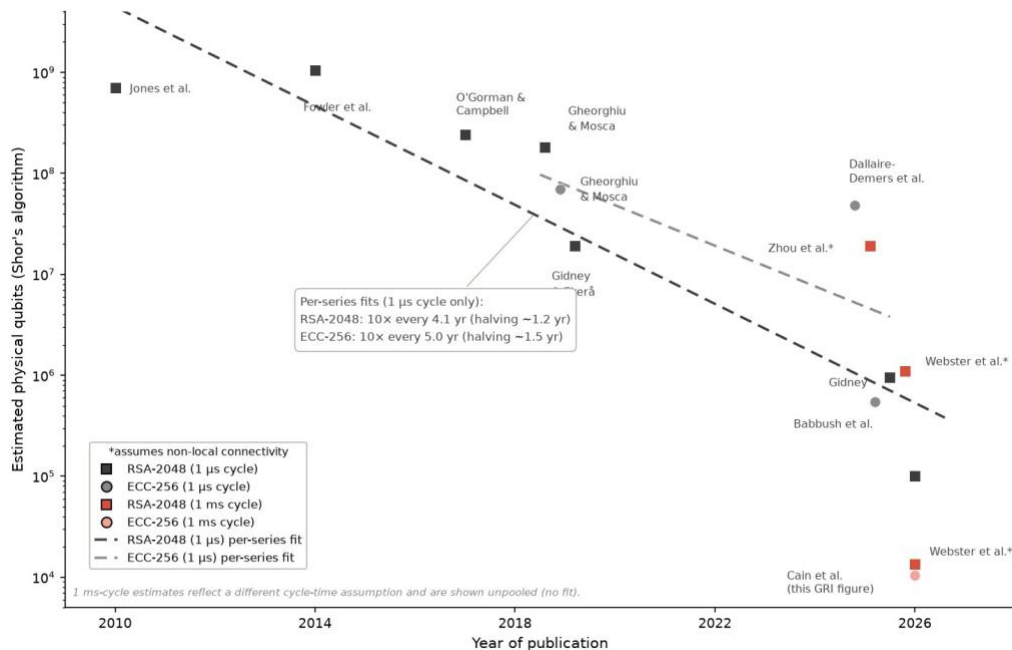


Figure 5: Exponential Decay Fit of Global Risk Institute Estimated Number of Qubits for Shor's Algorithm. Plotted qubit estimates are approximate reads pending the Global Risk Institute source table; the fit is reported per homogeneous series (RSA-2048 and ECC-256), not as a single pooled decay

This graph includes representations of some of these significant advancements in the first half of 2026 in reducing the estimated number of qubits needed for CRQC:

3.1.2.1 Pinnacle. The Iceberg Quantum Pinnacle Architecture has demonstrated reduction in qubit count against RSA-2048 (2048-bit RSA integers can be factored with less than 100,000 physical qubits). [84]

3.1.2.2 CalTech/IQIM. Caltech/IQIM QLDPC codes used for reconfigurable atom qubits using Shor's Algorithm against ECC-256 and RSA-2048 (Shor's algorithm possible with as few as 10,000 reconfigurable atomic qubits). [19]

3.1.2.3 Inria/CNRS. Inria/CNRS approach for reducing qubits against ECC (only 1,193 logical qubits required for ECC-256, roughly 42% fewer than the comparable RSA estimate). [85]

3.1.2.4 Hansung/Nanyang. Hansung/Nanyang ECDLP approach with an alternative approach to the computation of point multiplication in Shor's algorithm for improved quantum circuits used against ECC (up to a 40% improvement in the product of qubit count and circuit depth). [86]

3.1.2.5 QuEra. Startup quantum computing company promising thousands of logical qubits with a logical error rate at the 7-sigma level by 2028 to 2029 by cutting the overhead to 20 hardware qubits per logical qubit. [87]

3.1.3 PQC Vulnerability Spectrum for Lattice-Based PQC

Given the challenges of the current estimates of a timeline of at least a decade for migration to new encryption solutions, it is understandable why NIST has not made any formal predictions for CRQC threat risk for lattice-based PQC or for a migration timeline to the next generation of cryptography beyond PQC. At various points in the initial proposal for and review of proposed PQC algorithms, the impressions from the NIST proposals, presentations and reports can be interpreted as a general desire that the proposed PQC encryption algorithms would be quantum-resilient for at least a couple of decades after approval. [88]

The focus of much of the external analysis of the proposed PQC encryption algorithms was centered around use of Shor’s and Grover’s algorithms implemented by quantum computers or hybrid quantum/classical computer attacks, and on the strength of lattice-based cryptography against these kinds of attacks. NIST confirmed this focus on the presumed strength of lattice-based cryptography in its recent Fourth Status Report which selected for evaluation a set of PQC algorithms that do not use lattice-based cryptography to continue in the approval process as a potential PQC backup algorithm. (“The security of ML-KEM (FIPS-203) is based primarily on the presumed hardness of certain computational problems in lattices.”) [89]

Unfortunately, what is missing from such a “presumed hardness” assumption for projecting the cryptographic strength of lattice-based cryptography based on the LWE hard problem is any consideration or evaluation of the impact on the computational problems posed by lattice-based solutions by new or improved quantum algorithms or quantum computing technologies, particularly where such improvements are undertaken with the assistance of AI and AGI. Recent papers suggest the use of modular arithmetic to boost the performance of ML models in attacks on LWE. [90] [91] The presumed hardness assumption is also suspect based on two recent surveys of proofs for PQC schemes which found that PQC security proofs suffer a larger reduction loss as compared to classical ECC/RSA security proofs, which negatively affects overall security. [92] [93]

To anchor these qualitative concerns in the quantitative security model on which the standardized parameters rest, the best-known attacks on LWE proceed by lattice reduction (the BKZ algorithm) with a block size β , whose cost is captured by the core-SVP model:

$$\text{cost} \approx 2^{c\beta}, c = 0.292 \text{ (classical sieving)}, c = 0.265 \text{ (quantum sieving)}$$

The required block size β is fixed by the scheme parameters through the root-Hermite factor $\delta(\beta)$:

$$\delta(\beta) = [(\pi\beta)^{1/\beta} \cdot \beta / (2\pi e)]^{1/(2(\beta-1))}$$

with the primal attack succeeding approximately when

$$\sqrt{\beta} \cdot \sigma \lesssim \delta(\beta)^{2\beta-d-1} \cdot q_{m/d}$$

where d is the lattice dimension, q the modulus, and σ the noise width. This is a simplified primaluSVP heuristic (with d the full embedded lattice dimension); an exact estimator additionally carries lower-order dimension and error-norm factors that do not materially change the block sizes reported here (the -1 in the exponent scales the right-hand side by $\delta^{-1} \approx 0.998$ at $\beta \approx 877$, shifting the required block size by less than one unit). For the Kyber-1024 parameter set this yields $\beta \approx 877$, giving an estimated classical cost near 2^{256} and a quantum cost near 2^{232} operations. These figures are the published basis for the scheme’s claimed security, and any argument that the scheme falls earlier must show concretely how one of these two quantities, the exponent c or the block size β , is reduced. Section 4.5 develops exactly that accounting.

Two caveats temper these figures, and both cut against an earlier break rather than toward one. First, the quantum exponent $c = 0.265$ assumes Grover-amplified sieving with idealized, low-cost quantum random access memory (QRAM); once realistic memory-access and circuit-depth costs are charged, recent analyses indicate that the quantum advantage over classical sieving narrows substantially and may largely vanish, so 0.265 is optimistic for the attacker and the classical 0.292 may be closer to the operative exponent. Second, the core-SVP model counts only a single shortest-vector call and omits the additional reduction calls and overhead that a full attack incurs, so the values near 2^{256} and 2^{232} are conservative lower bounds and the true work factor is higher. Both points widen the margin that any earlier-break argument must overcome.

The current best quantum sieving time exponent, $2^{0.2563 \cdot d}$ (Bonnetain, Chailloux, Schrottenloher and Shen, EUROCRYPT 2023), was achieved by human cryptanalysts and published before the assumed 2028 AGI onset. Erosion from 0.265 to 0.2563 has therefore already occurred without AGI. This narrows the room left for AGI-driven erosion, supporting the contention that the collapse parameter must carry any break, and it rebuts the objection that the model gives human cryptanalysis too little credit. The same literature that fixes the list size (Cho, Hhan, Kim, Lee and Shen, IACR ePrint 2024/1700) also achieves $2^{0.2653 \cdot d}$ time with QRAM reduced to $2^{0.05778 \cdot d}$, which cuts against the QRAM caveat; the same source proves a lower bound of $2^{0.2925 \cdot d - 2^s}$ at QRAM size 2^s , with no quantum speedup available without QRAM under its assumptions, which cuts back in its favor. Both readings belong in the caveat, or it reads as selective.

3.2 Survey of Potential Improvements to Quantum and AI Technologies

Given the rapid expansion in both the capability and reach of AI over the last few years, and the even greater potential of AGI in the next few years, the biggest opportunities for potential improvements in quantum algorithms or quantum computing technologies are likely to be driven by the application of AI and then AGI to the design, optimization and innovation of quantum algorithms and computing technologies.

This section will briefly highlight some of the potential improvements related to CRQC+AI, as well as other quantum and AI related technologies that support the high-level presentation of these in Figure 3. The AI-capability results and quantum-hardware advances surveyed here are cited as evidence of the pace of AI-assisted progress and as enabling technologies; none of them constitutes a demonstrated cryptanalytic advance against standardized post-quantum cryptography.

3.2.1 AI Improvements for CRQC+AI

Some very recent examples of the unexpected advancements and innovations achieved by these new AI models include:

3.2.1.1 Using AI to Improve AI Models

Developments in AI models have reached an inflection point as of 2025, where AI models and constructs are now being improved upon by AI. [94]

3.2.1.2 Claude Mythos AI Models

Anthropic has introduced a new class of frontier models known as Mythos. These models are notably capable at discovering software vulnerabilities: independent reporting describes the Mythos-class models as particularly adept at detecting software flaws, including some that had gone undiscovered for years. [95] [96] Anthropic initially limited the availability of these models to a small set of organizations as part of its Project Glasswing. [97] A more broadly available variant, Fable 5, was released with specific safeguards intended to reduce the likelihood of misuse for cybersecurity tasks, and became available again, after a temporary suspension, as of July 1, 2026. [98] A systems-engineering

analysis has further examined the implications of such frontier-class models for post-quantum cryptographic migration. [99]

3.2.1.3 OpenAI Math Proof AI Models OpenAI recently announced that one of its generalpurpose reasoning models, not a system trained specifically for mathematics, disproved an 80-yearold conjecture in discrete geometry concerning the planar unit distance problem first posed by Erdős in 1946. [100]

3.2.2 Quantum Computing Improvements for CRQC+AI

IBM and Quantinuum have updates on the current state of quantum computing technology and roadmaps for future quantum computing developments that currently project hundreds to thousands of logical qubits capable of running billions of quantum gates. [101] [102] A recent announcement by Amazon and QuEra projected useful, error-correcting quantum computing by 2028. [103] Numerous other related and complementary improvements could further enhance the capabilities of quantum computing. The advances surveyed below are enabling technologies whose bearing on practical cryptanalysis is mostly indirect, and several remain early-stage research rather than near-term threats, as discussed for Rose's Law below. These include:

3.2.2.1 Qudits/Qutrits. Beyond qubits, which resolve to just two states upon measurement, work is progressing on the development of qudits and qutrits, which can resolve to m -dimensional states upon measurement, with the possibility of n entangled qudits representing m^n possible combinations. [104]

3.2.2.2 Noise/Heat Reductions. Work on using quantum information is also advancing, including use for hyper-entanglement of neutral atoms using a novel form of cooling to detect and correct thermal motion excitations, [105] and use of mirrored reflections to eliminate quantum noise. [106] **3.2.2.3 Quantum Switch** Cisco has recently announced a Universal Quantum Switch, a working research prototype that routes quantum information between quantum processors by entanglement at room temperature. [107]

3.2.2.4 Synthetic Dimensions for Photonic/Magnon/Quantum States. Various organizations are developing enhanced methods for manipulating photonic/magnon/quantum states to enhance coherence, entanglement and quantum information processing by using synthetic dimensions and quantum entanglement. [108] [109] [110] [111] [112] [113] [114] [171]

3.2.2.5 Quantum Circuits for SHA-1. Hansung University has developed a SHA-1 quantum circuit with a minimized circuit depth. [115] IBM has demonstrated a Quantum Fourier Transform (QFT) on its Quantum Heron processor. [116]

3.2.2.6 Quantum Hardware/Error Correction. Most of the recent efforts on exploring the relationship between AI and quantum computing have focused on using AI to enhance quantum computing hardware and error correction and quantum computing speedups. [117] [118] [119] [120] [121] [122]

3.2.2.7 Rose's Law. Just as Moore's Law projects the accelerating growth of classical computer technologies, [123] Rose's Law makes a similar projection for quantum computing technologies. [124] The natural human assumption that the spectrum of threats posed by CRQC+AI can safely be assumed to be limited to merely linear growth is a fallacious assumption that cannot be the basis for ignoring PQC

transition mandates and the need for crypto-agility to address these exponentially expanding threat profiles.

Stated quantitatively, Rose's Law is an exponential, not linear, growth model for usable quantum capability:

$$Q(t) = Q_0 \cdot 2^{(t - t_0)/\tau}$$

where Q_0 is the capability at a reference year t_0 and τ is the doubling time. This is the growth that feeds the hardware budget of Section 4.5: because that budget is the base-2 logarithm of the usable operations an attacker can afford, an exponential $Q(t)$ yields a budget $B(t)$ that rises linearly in t , which is exactly the constant-doubling form assumed there. Treating the threat as linear understates it by the difference between t and 2^t .

One qualification is essential here. Rose's Law is an empirical observation about the qubit counts of quantum annealers (the D-Wave line), and annealer qubits are neither gate-model qubits nor fault-tolerant logical qubits; they do not by themselves bound the resources available for Shor, Grover, or lattice sieving. The hardware budget $B(t)$ of Section 4.5 is therefore properly anchored to fault-tolerant logical-qubit and error-rate roadmaps for gate-model machines, with Rose's-Law scaling retained only as a loose upper envelope on the pace of progress, not as a direct count of cryptanalytically usable qubits.

The hardware driver in this paper is deliberately platform-agnostic in the same sense: it keys on effective error-corrected capability rather than on the roadmap of any single modality. Superconducting programs (IBM, Google), trapped-ion systems (Quantinuum, IonQ), photonic architectures (PsiQuantum), and neutral-atom platforms differ in qubit counts, error rates, clock speeds, and scaling cadence, and none is singled out by the model; what enters as the driver is the aggregate pace at which fault-tolerant logical capacity becomes available, from whichever platform or combination delivers it.

3.2.3 Quantum Algorithm Improvements for CRQC+AI

Some of the work of CRQC+AI relative to CRQCs has focused on optimizing and/or hacking both classical and PQC cryptography, including the use of CRQC+AI to enhance lattice reduction algorithms. [125] Current estimates based on topology classification are in the range of 100-200 different kinds of quantum algorithms. [126] Given the complexity of both quantum algorithms and programming quantum computers, it is likely that the various kinds of AI/ML/DL learning models and tools may be substantially faster at exploring variations and even completely different approaches to design and optimization of quantum programming of quantum algorithms in ways that humans might not even consider.

3.2.3.1 Language Models Examples of various AI models potentially useful for improving lattice reduction algorithms and other CRQC relevant quantum algorithms include language model programs and dynamic word embedding. [127] [128]

3.2.3.2 Quantum Algorithms Reinforcement learning has already discovered faster classical algorithms, for example improved sorting routines found by deep reinforcement learning, [129] which is direct evidence that AI can autonomously surpass human-designed algorithms. The same approach is expected to yield new kinds of quantum algorithms beyond the HHL and low-dimensional geometric algebra approaches described in this paper, including better approaches to reducing optimization problems via decoded quantum interferometry [130] and traveling salesman solutions [131], and solutions to quantum permutation puzzles. [132] Contests for using quantum algorithms to break BitCoin encryption are examples of how these new kinds of quantum algorithms are being encouraged. [133]

3.2.4 Side-Channel Attacks

Numerous types of new side-channel attacks are being developed for both classical and PQC algorithms using CRQC+AI. [134] These include attacks such as practical fault attacks on randomness, [135] single-trace passive side-channel attacks on HQC, [136] first-order masked ML-KEM attack using multiple distinct chosen ciphertexts, [137] and exploitation of noisy single-bit leakage. [138]

The model does not separately price implementation or side-channel compromise, that is, trace-acquisition cost, success probability, or countermeasure state. On any reasonable assessment, implementation compromise of a deployed PQC stack before 2035 is more probable than a structural lattice break and requires no quantum computer.

3.2.5 Key Recovery Attacks

Numerous types of new key attacks are being developed for both classical and PQC algorithms using CRQC+AI. These include attacks such as neighborhood search attacks on Oracle MLWE for MLKEM, [139] data repetition and stepwise regression, [140] quantum truncated differential attacks, [141] multi-instance key degradation, [142] as well as hybrid-adaptive-LDPC attacks on ML-KEM. [143]

3.2.6 Hybrid Attacks

Numerous types of hybrid attacks are being developed for both classical and PQC algorithms using CRQC+AI. For PQC algorithms, these include primal hybrid attacks against sparse LWE [144] [145], Polynomial LWE (PLWE) [93], Ring-LWE [146], and dual attacks [147].

3.3 An Acceleration-Factor Model for AI-Compressed Timelines

The upper-bound scenarios in Section 3.1 assume that AI compresses the calendar time to a CRQC, but that assumption does most of the work in the aggressive projections and deserves to be stated as an explicit, falsifiable model rather than a qualitative claim. This section sketches such a model. Its purpose is not to make the timeline more aggressive but to make whatever timeline it produces auditable: every input is a named quantity with a baseline anchored in published progress, an acceleration factor anchored in observed machine-learning results, and a stated condition under which the projection would be revised. Because every quantity is named and anchored, the model is built to be rerun: any reader can vary the parameters, regenerate the curves, and compare successive runs against the milestones that have since become historical data points, in the same way that published quantum resource estimates and expert-elicitation surveys are periodically revised. The model deliberately holds the quantum algorithm fixed. AI is allowed to accelerate engineering and design; it is not allowed to change the complexity of Shor's algorithm or to lower the fault-tolerance threshold.

3.3.1 What the Acceleration Factor Scales, and What It Does Not

The single most important boundary in this model is between what AI can plausibly accelerate and what it cannot. AI is permitted to compress the engineering and design loop: decoder design and real-time decoding, pulse-level gate calibration and optimal control, fabrication and materials search, error-correcting-code discovery, and circuit compilation and optimization. AI is not permitted to alter the underlying complexity theory or physics: it does not change the polynomial scaling of Shor's algorithm, it does not lower the fault-tolerant error-correction threshold, and it does not repeal qubit coherence limits.

This separation is what keeps the model honest. A projection in which AI accelerates everything uniformly is indistinguishable from assuming the conclusion; a projection in which AI accelerates only the identifiable engineering bottlenecks can be checked channel by channel against the published record.

3.3.2 Four Channels and Their Baselines

We decompose progress toward a CRQC into four engineering channels, each with its own baseline rate of improvement and its own acceleration factor. Treating them separately, rather than as a single global rate, is what lets each factor be argued and falsified independently. The channels, their published baselines, and the engineering tasks where AI plausibly contributes are summarized in Table 2 below. Each acceleration factor below is calibrated rather than assumed: its baseline rate is taken from a published progress trajectory and its value from a measured machine-learning speedup, with the supporting quantities given in the calibration Table 3 that follows. The factors are kept as ranges, low to central to high, to carry the uncertainty honestly, but they are no longer free parameters.

Magic-state distillation, a further cost of fault-tolerant Shor, is treated within the code-efficiency and logical-qubit channels rather than as a separate one. At the modeled binding point the raw logical-qubit count is the constraint (Table 2), and whether distillation would bind later than that count is not established; recent qLDPC codes and improved protocols have reduced its overhead. It is therefore not carried as an independent channel. Should a distillation demonstration change that assessment, the tracking indicators of Section 3.3.5 promote it to its own factor rather than revising the whole projection at once.

Channel	Pre-AI baseline trajectory (anchor)	What AI plausibly accelerates	Acceleration factor A_i (low / central / high)
Physical qubit count and modular interconnect	Vendor roadmaps; for example, on the order of 200 logical qubits by 2029 and 1,000 or more by the early 2030s [148]	Fabrication and yield search, module layout and packaging, calibration at scale	$A_1 = 1.0 / 1.3 / 2.0$
Two-qubit gate fidelity and physical error rate	Below-threshold physical error rates demonstrated on current processors [149]	Pulse-level optimal control, calibration, crosstalk mitigation	$A_2 = 1.0 / 1.5 / 2.5$
Logical error rate per cycle and code efficiency Λ	$\Lambda \approx 2.14$ per two units of code distance; 0.143% per cycle at distance 7 [149]	Decoder accuracy (learned decoders [122]) and error-correcting-code discovery	$A_3 = 1.0 / 1.4 / 2.0$
Code-cycle time and realtime decoder latency	Real-time decoder latency on the order of 60 μ s; cycle time $\approx 1 \mu$ s [149]	Faster learned decoders and real-time inference on FPGA or ASIC	$A_4 = 1.0 / 1.3 / 1.8$

Table 2: Engineering channels toward a CRQC, with their pre-AI baseline trajectories and the machine-learning-anchored acceleration factors.

The required-improvement factor for the qubit-count channel is fixed by the resource estimate for the target attack, not chosen freely. Factoring RSA-2048 was estimated in 2019 at roughly twenty million physical qubits running for about eight hours, [79] and revised in 2025 to under one million qubits running for under a week. [82] The qubit-count target has therefore moved by more than an order of magnitude

in six years through algorithmic and error-correction advances rather than hardware alone, and the model treats that moving target explicitly rather than assuming a fixed endpoint.

Setting $t_0 = 2026$, the required factors and baseline rates, with their anchors, give the following per-channel calibration in Table 3. The time each channel needs at no acceleration is $\tau_i = \ln(F_i) / r_i$; the latest of these is the binding constraint.

Channel	Required factor F_i (anchor)	Baseline rate r_i , e-folds per year (anchor)	τ_i at $A_i = 1$ (years)	Implied year at $A_i = 1$
Logical qubit count (binding)	$\approx 10^3$ (order-1 today to $\approx 1,400$ logical for RSA-2048 [82])	1.1 (IBM roadmap: order-1 to $\approx 2,000$ logical by 2033 [148])	6.3	≈ 2032
Gate fidelity and physical error rate	≈ 3 (few $\times 10^{-3}$ today to 10^{-3} target [149], [82])	0.4 ($\approx 10\times$ error reduction over ≈ 6 years; 99% to 99.9% [149])	2.7	≈ 2029
Code efficiency and overhead	≈ 3 (qLDPC overhead headroom of order $10\times$ [148])	0.4 (recent $\approx 90\%$ overhead reduction via qLDPC [148])	2.7	≈ 2029
Code-cycle time and decoder latency	≈ 60 ($\approx 60 \mu s$ to $1 \mu s$ [149])	1.0 (Relay-BP 5 to 10 times faster [148]; learned decoders [122])	4.1	≈ 2030

Table 3: Per-channel required factors, baseline rates, and the implied CRQC readiness year under baseline progress (acceleration factor of one).

3.3.3 Empirical Anchors for the Acceleration Factors

Each acceleration factor should be calibrated to an observed machine-learning result rather than chosen to produce a target date. For the decoder and logical-error channel, a neural-network decoder has reported on the order of six percent fewer errors than the best slow decoder and roughly thirty percent fewer than the best real-time decoder on the same hardware, [122] a measured improvement that bounds the near-term contribution of learned decoding. For the logical-error and code-cycle channels, the demonstration of below-threshold surface-code operation, with a logical-error suppression factor of about 2.14 per two units of code distance and a real-time decoder latency on the order of sixty microseconds, [149] sets the baseline trajectory those factors accelerate. Vendor roadmaps targeting on the order of two hundred logical qubits by 2029 and a thousand or more in the early 2030s [148] anchor the qubit-count channel. Where no measured machine-learning speedup exists for a channel, its acceleration factor should default to one, that is, no acceleration, rather than to an optimistic guess.

3.3.4 From Channels to a Distribution Over the Crossover Date

For each channel let F_i be the factor by which that channel must still improve to reach the CRQC target, r_i the baseline rate of improvement (in e-folds per year) drawn from the published trajectory, and A_i the acceleration factor. The calendar time the channel requires is then $\tau_i = \ln(F_i) / (A_i \cdot r_i)$

and, because a CRQC requires every channel to reach its target, the crossover date is set by the slowest, or binding, channel:

$$T = t_0 + \max_i \tau_i$$

Treating the A_i , and where appropriate the F_i and r_i , as ranges rather than point values turns this into a distribution over T rather than a single date. The output of interest is then not a single Q-Day but a probability that a cryptographically relevant capability arrives before a given year, $P(T \leq \text{year})$, which is exactly the quantity Mosca's inequality needs for Z : any nonzero probability mass inside the secrecy horizon X is sufficient to make migration rational today. This formulation also makes the bounds of Section 3.1 explicit: the linear lower bound corresponds to every A_i set to one, and the exponential upper bound corresponds to the high end of the acceleration ranges combined with the most aggressive resource estimates.

As a worked example, take $t_0 = 2026$ and the figures in the calibration table. At no acceleration the four channels reach their targets in roughly 2029, 2029, 2030, and 2032, so the binding channel is the raw logical-qubit count, and the no-acceleration crossover is about 2032. Applying the acceleration range to that binding channel, $T = 2026 + \ln(10^3) / (A_1 \cdot 1.1)$, gives about 2031 at the central factor $A_1 = 1.3$ and about 2029 at the high factor $A_1 = 2.0$. These outputs, roughly 2032, 2031, and 2029, reproduce the RSA-2048 crossovers shown in Figure 9, where the no-AGI Track 0 crosses by about 2032 on hardware scaling alone and the aggressive Track 3 crosses by about 2030, and they bracket IBM's own internal 2029 target. [148] The acceleration-factor model is therefore not an independent or more aggressive forecast; it is the mechanism that produces the same spread of dates the figures show, expressed in auditable parameters.

This decomposition is the granular form of the aggregate Rose's-Law hardware growth of Section 3.2 and the budget $B(t)$ of Section 4.5, and the two are calibrated to agree: the binding-channel crossover of about 2032 at no acceleration coincides with the Track 0 hardware-only RSA crossover the aggregate model produces. The engineering acceleration factors A_i are also distinct from the research-acceleration factor R of Section 4.5 and are not interchangeable with it. The A_i compress the time to build a fault-tolerant machine, the hardware track that drives the RSA and ECC crossovers, whereas R compresses cryptanalytic research effort, the software track on which the lattice and hash-based contingencies of Section 4.5 separately depend; one set of numbers governs hardware and the other governs cryptanalysis, and the timeline uses both rather than a single blended rate. Two simplifications should be stated plainly. The per-channel rates r_i are averages over the published roadmap horizon and are in practice front-loaded, with the steep early logical-qubit ramp expected to slow in later years. And channels one and three are physically coupled, since better codes lower the qubit-count target, so treating them as independent is conservative for the binding channel rather than optimistic.

Two features of this calibration are worth stating plainly. First, the binding channel is the one where AI has the least direct evidence of acceleration: raw qubit-count scaling. The strongest measured machine-learning contributions, in decoding, calibration, and code design, fall in the non-binding channels, which already reach their targets earlier and so cannot move the crossover on their own. Second, the route by which those evidenced contributions do move the date is indirect but real: better codes, higher fidelity, and faster decoding lower the qubit-count target F_1 , which is exactly the mechanism behind the fall of the RSA-2048 estimate from twenty million to under one million qubits between 2019 and 2025. [79], [82] The aggressive end of the timeline therefore depends less on AI building qubits faster than on AI continuing to shrink the number of qubits required, a dependency this model makes explicit rather than absorbing into a single factor.

3.3.5 Falsification Conditions and Tracking Indicators

Because the model is built from named quantities, it can be falsified or updated by observation rather than argued in the abstract. The following indicators, tracked over the next eighteen to thirty-six months, could move the projected crossover date earlier or later:

- physical qubit counts and modular interconnect scaling meeting, beating, or missing the published vendor roadmap milestones;
- logical error rate per cycle continuing to fall by a factor of two or more per two units of code distance, or instead stalling above the level the target computation requires;
- real-time decoder latency reaching and holding the microsecond regime at increasing code distance, or failing to keep pace with the cycle time;
- magic-state production and injection moving from theory to a first hardware factory demonstration, or remaining undemonstrated;
- the published resource estimate for the target attack continuing to fall, as it did from twenty million to under one million qubits, or instead stabilizing.

Each indicator maps to one of the four channels, so an observation that beats or misses a milestone updates a specific acceleration factor rather than the whole projection at once. This converts the timeline from a single prediction into a tracker, consistent with the spectrum-of-risk-dates framing adopted in Section 3.1.

These indicators define the model's explicit update mechanics. The intended protocol includes the following:

- a periodic, at least annual, recalibration cycle:
- each observed milestone is mapped to its channel;
- the corresponding required factor F_i , baseline rate r_i , or acceleration factor A_i is re-estimated from the new anchor; the distribution over T is recomputed; and
- the revised parameter set is published as a versioned rerun alongside its predecessors, so that conformance to, or deviation from, the previously modeled curves is itself preserved as calibration data.

Over successive update cycles this procedure narrows the acceleration-factor ranges toward what the accumulating historical record supports, in the same spirit in which expert-elicitation surveys are re-fielded and published resource estimates are revised. This periodic update protocol is part of what qualifies the model as an estimation instrument that improves with observation rather than a static projection.

Finally, the model makes the paper's central claim explicit and bounded: it accelerates the arrival of fault-tolerant Shor's algorithm, the one method this paper does not rule out, rather than creating a new near-term attack. The methods evaluated in Sections 2.3 and 2.4, quantum annealing and the variational quantum eigensolver, do not contribute a standalone break, and the linear-algebra and geometric-algebra approaches of Sections 2.1 and 2.2 contribute at most polynomial assistance or remain exploratory. The novel content of the timeline is therefore concentrated in the acceleration factors above, which is precisely why they are stated as named, anchored, and falsifiable quantities, rather than being folded into a single qualitative assumption.

Reproducibility versus validation. Appendix A permits exact regeneration of every curve from the stated equations and seeds; that is reproducibility, and it is distinct from validation. Three levels of validation should be kept apart. First, outcome validation, whether reality reaches a modeled crossover, is prospective: the earliest testable event is the RSA-2048 crossing near 2030, so as of July 2026 no projection has yet been checked against its own predicted event. Second, concordance validation is available now and is reported in Table 4: the model's crossover ranges are consistent with five independent expert forecasts, standards timelines and vendor roadmaps (Mosca and the Global Risk Institute, NIST IR 8547, CNSA 2.0, IBM, and Google/Gidney), which is convergent support rather than proof. Third, leading-indicator validation is also available now: the five tracking indicators of this section, observable over the next eighteen to thirty-six months and each mapped to a specific channel, give early corroborating or falsifying signal well before 2030, and one of them, the fall in the published resource estimate from roughly twenty million to under one million qubits, is a retrodictive anchor the model's trend already matches. The annual recalibration protocol above turns these into a standing check. Accordingly, the auditability and reproducibility claims describe repeatability; outcome correctness remains open, to be established by the tracking protocol as the record accumulates. A second, like-for-like retrodiction is developed in Section 4.5: the observed decline of the quantum sieving exponent runs about seven times too slow to reach the sieving floor by 2033, independent historical support for the contingency on the collapse parameter.

4. Using AI to Project the Advancement of CRQC+AI Vulnerabilities

The rapid rise in the capability and reach of AI in recent years has opened previously unforeseeable opportunities for improvements in quantum algorithms and quantum computing technologies. Both AI-enhanced CRQCs and AGI-enhanced CRQCs are expected to advance the design, optimization and innovation of quantum algorithms and computing technologies. This section includes the results of using the highest ranked currently available AI tools to render and visualize the CRQC+AI vulnerability timelines from author-specified parameters and equations (Anthropic Claude Table 5). [150] These author-parameterized, AI-rendered charts (all modeling assumptions, parameters, and calculations author-controlled; see the provenance statement at the end of this section) evaluate four different scenarios with respect to the risk profile for cracking classical asymmetric RSA/ECC public key encryptions and PQC approved asymmetric encryption (FIPS 203 and FIPS 204), with two additional algorithms evaluated in Appendix C and not in the body of this paper, PQC asymmetric encryption (FIPS 205), as well as AES symmetric encryption (FIPS 197). [151] [152]

The end-of-2028 horizon used in these scenarios reflects publicly stated developer timelines rather than an assumption unique to this paper. Anthropic's founder and chief executive has projected powerful AI on a 2026 to 2027 timeline, [153] [154] and its submission to the United States Office of Science and Technology Policy places such systems within the current administration term, that is, before the end of 2028. [155] These developer timelines are echoed across the field by forecasts that bracket the 2028 horizon rather than uniformly confirming it: Google DeepMind's Chief AGI Scientist has for over a decade assigned roughly even odds to AGI by 2028, [156] DeepMind's chief executive gives a somewhat longer five-to-ten-year horizon, [157] the chief executive of Microsoft AI expects AI to reach human-level performance on most professional, computer-based tasks within roughly 12 to 18 months for a professional-task horizon near 2027 rather than full general intelligence, [158] and a former OpenAI researcher argues AGI is plausible by 2027. [159] A prominent dissent holds that current large language models are not on a path to human-level intelligence and that AGI would require different architectures on a longer horizon. [160] Such forecasts come from interested parties and remain contested, and the horizon is adopted here only as a scenario input rather than a settled date; Track 0 accordingly retains a

fully no-AGI baseline for readers who do not accept the premise. The four different scenarios shown in each chart track the risk threats for breaking the relevant encryption algorithms under different assumptions based on AGI being available for CRQC by the end of 2028:

- Track 0: No-AGI baseline (zero AGI-gated software capability).
- Track 1: Low AGI-driven software capability.
- Track 2: Moderate AGI-driven software capability.
- Track 3: High AGI-driven software capability (most aggressive regime).

These four scenario tracks are distinct from the four engineering channels of Section 3.3: the channels decompose engineering progress toward a CRQC, whereas the tracks grade the intensity of the AGI-gated software driver (its rate r_k and ceiling B_k in Appendix A). The four tracks share one functional form and differ only in that intensity, so the labels above denote capability level rather than distinct attack mechanisms. The runtime and feasibility curves in the figures that follow are produced by the combined hardware and software capability model documented in full, with parameters and pseudocode, in Appendix A; the cost model of Section 4.5 provides a separate analytical restatement in explicit cryptanalytic units; because it shares the same effective-dimension-collapse conjecture, it is a consistency check rather than independent corroboration.

The curves in Figures 6A/6B through 8A/8B, together with the Appendix C stress-case Figures C1A/C1B and C2A/C2B, are generated from a single parameterized capability model rather than from individual expert or analyst projections. Cryptanalytic capability is expressed as the accumulated orders-of-magnitude reduction in attack runtime over time and is modeled as the sum of two exponential drivers: a hardware driver representing the accelerating growth of effective quantum resources (Rose's-Law scaling together with the qubit-count reductions surveyed in Section 3), that is present in every track because hardware progress is not contingent on AGI; and an AGI-gated software driver representing AI-assisted advances in cryptanalysis, which differs by track. The full specification, with parameters, calibration, and pseudocode, is given in Appendix A. One reading convention applies to all these figures: the vertical runtime axis is a normalized index, not a per-scheme physical estimate. Every scheme starts from the same baseline value R_0 (Appendix A), so the curves support within-scheme trends and crossover timing, but differences between schemes on this axis do not represent their true differences in absolute attack cost, which span dozens of orders of magnitude and are accounted for in the cryptanalytic units of Section 4.5.

The Anthropic Claude Fable 5 AI model as of July 2026 was used as a drafting and visualization assistant for these charts and for an editorial review of the paper. All modeling assumptions, parameters, and calculations are author-controlled and reproducible from the equations and calibration tables given in Sections 3 and 4.

4.1 RSA-2048 Estimates of CRQC+AI Vulnerability: Mechanism-Backed Tier

Figures 6A and 6B evaluate the runtime attack and modeled feasibility estimates for breaking RSA2048 public key encryption under each of the different vulnerability tracks for the four different assumption scenarios. In TLS 1.3 specifically, RSA key transport was removed; RSA remains relevant for certificate authentication, where RSA signatures use RSASSA-PSS, and for legacy interoperability and downgrade exposure, while ephemeral elliptic-curve Diffie-Hellman (ECDHE) and ECDSA are the direct TLS 1.3 key-exchange and authentication mechanisms.

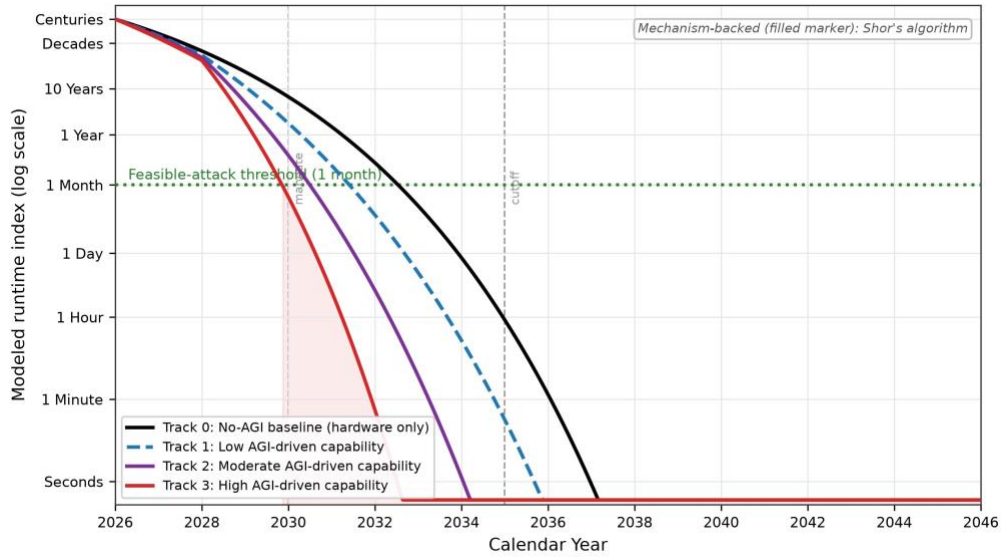


Figure 6A: Projected Attack Runtime Against RSA-2048 (classical RSA). Certificate Authentication / Legacy TLS Exposure

The vertical runtime axis is a within-scheme index on a common baseline (R0); values are not comparable across schemes (see Section 4).

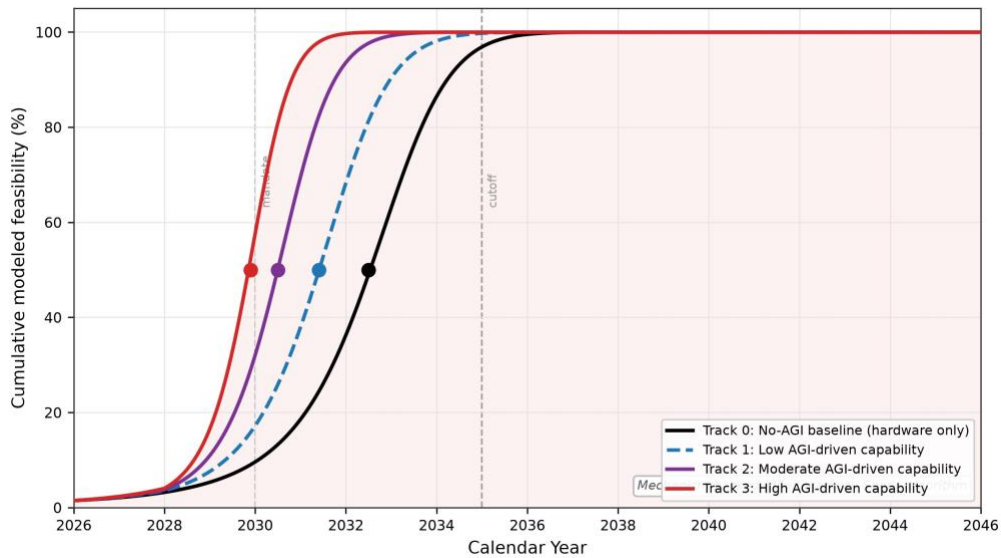


Figure 6B: Modeled Feasibility Estimates for RSA-2048 (classical RSA)
Certificate Authentication / Legacy TLS Exposure

4.2 ML-KEM Kyber-768/1024 Estimates of CRQC+AI Vulnerability (FIPS 203): Contingency Tier

Figures 7A/7B and 7C/7D evaluate the runtime attack and modeled feasibility estimates for breaking Kyber-768 and Kyber-1024 public key encryption (FIPS 203) under both the most common version of Kyber to be implemented under TLS (X25519+ ML-KEM Kyber-768) the hybrid X25519 + ML-KEM-768 group, which

major browsers enable by default and which Cloudflare reports as its widely deployed post-quantum key agreement, with over half of human-initiated traffic to Cloudflare now post-quantum encrypted, [71] and the version of Kyber required by the National Security Agency (NSA) under Commercial National Security Algorithm Suite 2.0 (CNSA 2.0), [164] under each of the different vulnerability tracks for the four different assumption scenarios.

Because TLS 1.3 runs on ML-KEM-768 (NIST category 3) in hybrid with classical ECDH (X25519) [161] and combines the two public key exchange results, an attacker must beat both the stronger parameter set of lattice-based ML-KEM and the classical ECDH component to derive the results of the key exchange. Figures 7A/7B evaluate the runtime attack and modeled feasibility estimates for breaking Kyber-768 in this hybrid context with ECDH X25519.

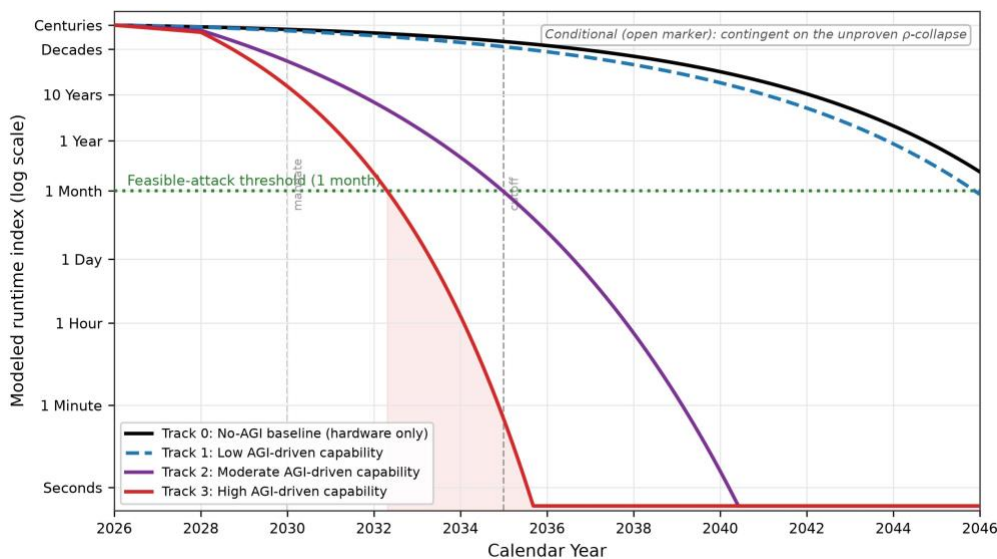


Figure 7A: Projected Attack Runtime Against Hybrid Kyber-768
ML-KEM and X25519 ECDH: Most Common TLS 1.3 Key Exchange for FIPS 203

*The vertical runtime axis is a within-scheme index on a common baseline (R0);
values are not comparable across schemes (see Section 4).*

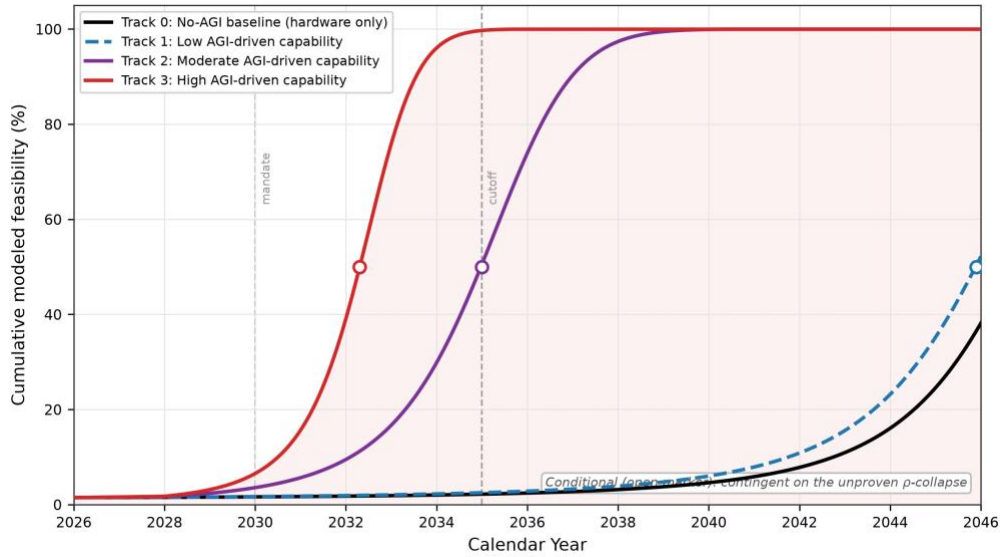


Figure 7B: Modeled Feasibility Estimates for Hybrid Kyber-768 ML-KEM and X25519 ECDH: Most Common TLS 1.3 Key Exchange for FIPS 203

Conditional scenario only: the Kyber-768 +X25519 curves assume an undiscovered effective-dimension-collapse mechanism of approximately $p \geq 0.52$. No such mechanism is currently known against standardized lattice schemes.

Figures 7C/7D evaluate the runtime attack and modeled feasibility estimates for breaking Kyber1024 public key encryption as a standalone (PQ-only) ML-KEM-1024 key agreement (NIST category 5), consistent with current IETF TLS Working Group Internet-Drafts. TLS 1.3 standardization currently includes separate Working Group drafts for hybrid and standalone post-quantum key agreement: the hybrid draft defines combinations such as SecP384r1MLKEM1024, [161] while the standalone draft defines ML-KEM-1024 as a PQ-only NamedGroup, [162] both still Internet-Drafts rather than final RFCs. This standalone configuration is relevant to CNSA 2.0-aligned deployments, in which NSA specifies ML-KEM-1024 (not ML-KEM-768); CNSA 2.0 should be read as specifying required quantum-resistant algorithms and strength levels for applicable high-assurance use cases rather than as a blanket prohibition on hybrid key establishment. [163] [164]

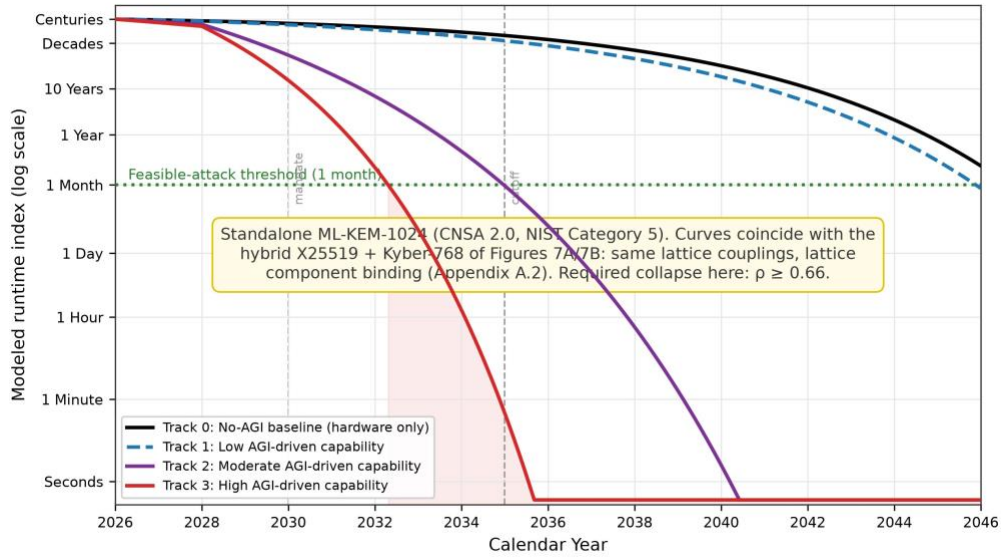


Figure 7C: Projected Attack Runtime Against Standalone Kyber-1024
ML-KEM-1024 (NIST Category 5): CNSA 2.0 TLS 1.3 Key Exchange for FIPS 203

*The vertical runtime axis is a within-scheme index on a common baseline (R0);
values are not comparable across schemes (see Section 4).*

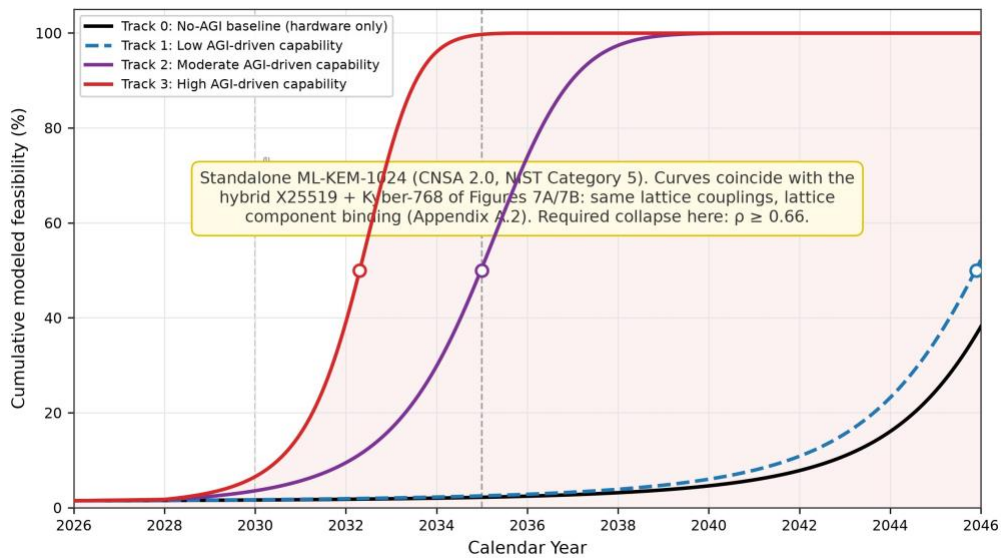


Figure 7D: Modeled Feasibility Estimates for Standalone Kyber-1024
ML-KEM-1024 (NIST Category 5): CNSA 2.0 TLS 1.3 Key Exchange for FIPS 203

Conditional scenario only: the Kyber-1024 standalone curves assume an undiscovered effective-dimension-collapse mechanism of approximately $\rho \geq 0.66$. No such mechanism is currently known against standardized lattice schemes.

4.3 ML-DSA-44 Estimates of CRQC+AI Vulnerability (FIPS 204): Contingency Tier

Figures 8A and 8B evaluate the runtime attack and modeled feasibility estimates for breaking MLDSA-44 (FIPS 204) under each of the different vulnerability tracks for the four different assumption scenarios.

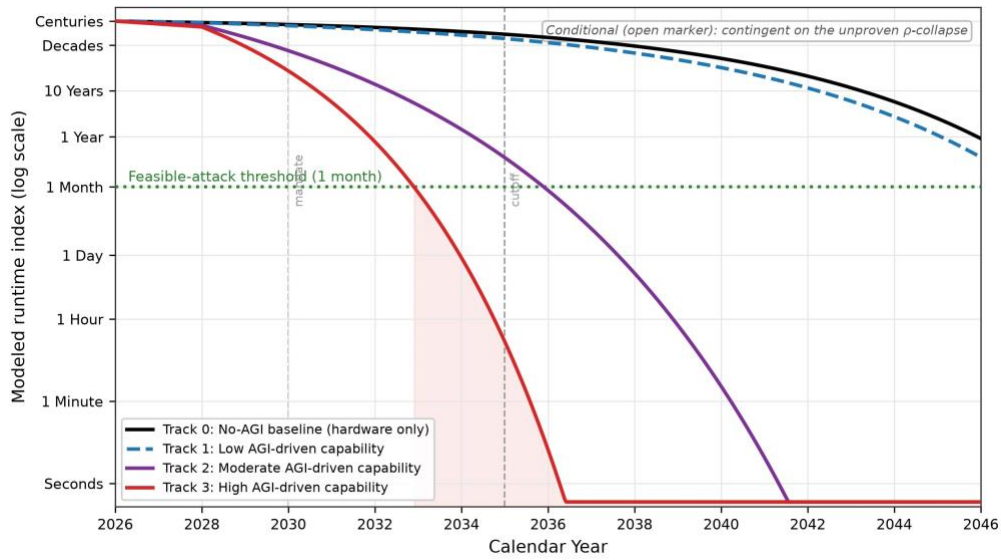


Figure 8A: Projected Attack Runtime Against ML-DSA-44 FIPS 204 TLS 1.3 Certificate Signatures

The vertical runtime axis is a within-scheme index on a common baseline (R_0); values are not comparable across schemes (see Section 4).

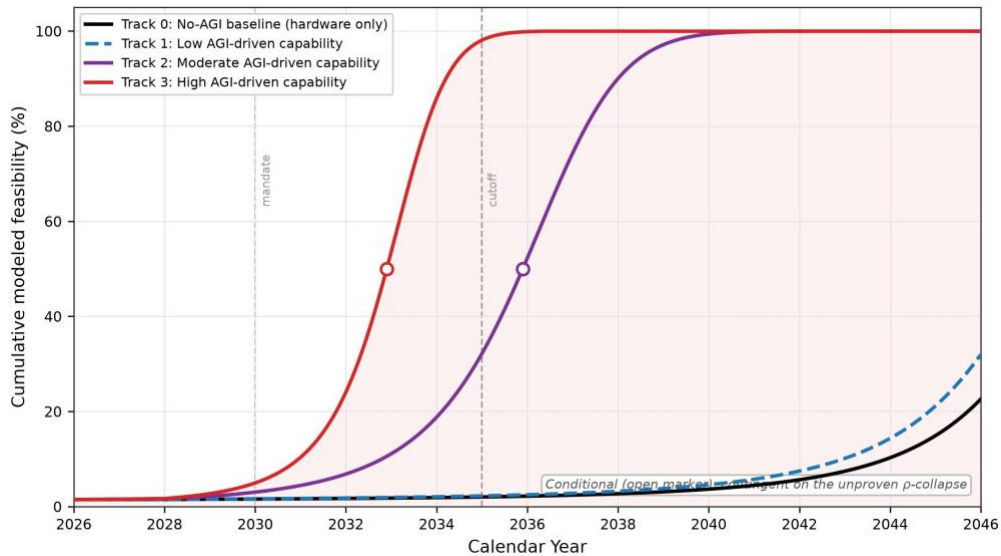


Figure 8B: Modeled Feasibility Estimates for ML-DSA-44 FIPS 204 TLS 1.3 Certificate Signatures

Conditional scenario only: the ML-DSA-44 curves assume an undiscovered effective-dimension-collapse mechanism of approximately $\rho \geq 0.17$. No such mechanism is currently known against standardized lattice schemes.

4.4 Estimates of CRQC+AI Vulnerability for SLH-DSA (FIPS 205) and AES-256 (FIPS 197)

The remaining two TLS 1.3 primitives, SLH-DSA (FIPS 205) and AES-256 (FIPS 197), show no mechanism-backed or contingency-backed exposure in this model. Their hypothesis-only stress cases, including runtime and feasibility Figures C1A/C1B and C2A/C2B, are presented in Appendix C so that the figures in the body of this paper remain confined to the two better-evidenced tiers.

4.5 Cryptanalytic Cost Model and the Dimension-Collapse Contingency The per-algorithm figures above are generated by the combined hardware and software capability model documented in Appendix A. This subsection is a separate analytical restatement of the lattice and hash-based tracks, not the generator of those figures: it restates the conditional break in explicit cryptanalytic units so that the single speculative assumption is isolated and quantified rather than hidden inside the shape of a curve. The two instruments agree by sharing the same conditional assumption, an undiscovered effective-dimension collapse, rather than by one being derived from the other. Building on the core-SVP cost of Section 3.1.3, the feasibility of a lattice break in calendar year t is expressed through three parameters, each carrying a concrete meaning:

- Δ (exponent erosion): an AI-discovered reduction of the quantum sieving exponent below its current value of 0.265, bounded below by a conjectured sieving floor so that $0.265 - \Delta \geq 0.2075$. This term captures faster algorithms but cannot, by itself, change the exponential form of the cost.
- This floor is a *resource* floor induced by the minimal sieve-list size, $|L| \approx 2^{0.2075 \cdot d}$ in the 2-sieve family (Kirshanova and Laarhoven, Lower Bounds on Lattice Sieving and Information Set Decoding, CRYPTO 2021, IACR ePrint 2021/785), not an optimality result on the time exponent. It is family-specific: 3-tuple sieving trades to $\approx 2^{0.1887 \cdot d}$ memory at $\approx 2^{0.2846 \cdot d}$ time, which does not change the argument directionally. The list must be produced and traversed, so it induces a floor on the operation count against which the budget $B(t)$ is measured.
- ρ (effective-dimension collapse): the payload of the geometric-algebra hypothesis of Section 2.2, modeled as a collapse of the core-SVP block size β (distinct from the capability-model coupling β of Appendix A), $\beta \rightarrow \beta(1 - \rho)$. It is the only speculative quantity in the model and has no known realization against random q -ary lattices; it is exposed deliberately so that the assumption can be examined.
- $B(t)$ (feasible attack budget): the base-2 logarithm of the quantum operations an adversary can afford in year t , increasing with the hardware driver (Rose’s-Law scaling and the qubit-count reductions surveyed in Section 3).

A scheme with core block size β becomes feasibly breakable in the first year for which the eroded, dimension-collapsed work factor falls within budget:

$$(0.265 - \Delta) \cdot \beta \cdot (1 - \rho) \leq B(t) \text{ [lattice break condition]}$$

The two drivers named in the model note above take explicit form. The hardware budget grows linearly in log space (a constant-doubling, Rose’s-Law form), while the software erosion is gated on the assumed AGI onset t_{AGI} (end of 2028):

$$B(t) = B_0 + r \cdot (t - t_0)$$

$$\Delta(t) = \Delta_{\max} \cdot (1 - e^{-R \cdot (t - t_{\text{AGI}})}) \text{ for } t \geq t_{\text{AGI}}, \Delta(t) = 0 \text{ otherwise}$$

Here R is a research-acceleration factor: the number of human-cryptanalyst-years of progress compressed into one calendar year by advanced AI. Track 0 sets $R = 0$ (no AGI); Tracks 1 and 2 raise R ; Track 3 is the most aggressive regime, implemented in Appendix A with the highest fixed track rate ($r_k = 1.30$), whose compounding produces the sharply convex capability accumulation visible in the Track 3 curves. The superintelligence premise is expressed through R rather than through any intelligence score, because a research-acceleration factor is measurable in principle (the quantum sieving exponent did fall over roughly fifteen years of human work, so a baseline rate exists to accelerate), whereas a normalized intelligence score has no defined meaning in the regime contemplated here. If a narrative anchor is wanted, an intelligence far beyond the human range is taken to drive R to large values; this anchor is explicitly illustrative, and no crossover year is derived from it.

For transparency, the two anchor values used in the worked examples below, $B(2033) = 80$ bits and $B(2036) \approx 92$ bits, determine the budget line completely: $B_0 = 52$ bits at $t_0 = 2026$ and $r = 4$ bits per year, equivalent to a doubling time of roughly three months in usable attack operations. This is an aggressive scenario input, reflecting compounded hardware, engineering, and orchestration gains under the CRQC+AI assumption, and it should be read as such rather than as a forecast of any vendor roadmap.

Worked examples, ML-KEM-768/1024 (Kyber-768 hybrid/1024 standalone, FIPS 203). For Kyber-1024 (standalone) with $\beta \approx 877$ and quantum exponent 0.265, take an already aggressive illustrative budget of $B(2033) = 80$ bits. The break condition becomes:

$$(0.265 - \Delta)(1 - \rho) \leq 80 / 877 \approx 0.091$$

Two limiting readings follow. With no geometric collapse ($\rho = 0$), the requirement $0.265 - \Delta \leq 0.091$ would force the sieving exponent below the conjectured floor of 0.2075; even at that floor the work factor is $2^{0.2075 \cdot 877} \approx 2^{182}$, still above the budget, so algorithmic erosion alone cannot reach 2033. With no exponent erosion ($\Delta = 0$), the requirement becomes:

$$(1 - \rho) \leq 0.091 / 0.265 \approx 0.344 \Rightarrow \rho \geq 0.66$$

In other words, under real complexity numbers a 2032 to 2035 Kyber-1024 break is mathematically contingent on the unproven geometric embedding reducing the effective lattice dimension by roughly two thirds. The model does not assert that such a collapse exists; it states precisely how large it would have to be. For the hybrid Kyber-768 + X25519 configuration that is the widely deployed post-quantum TLS default ($\beta \approx 625$), the same budget requires $\rho \geq 0.52$, and the classical X25519 component must additionally be broken.

A retrodiction against the historical record supports this reading. The best-known quantum sieving time exponent has fallen only from about 0.2653 in 2015 to 0.2563 in 2023 (Laarhoven; Chailloux and Loyer, ASIACRYPT 2021; Bonnetain, Chailloux, Schrottenloher and Shen, EUROCRYPT 2023), a human-progress rate near 1.2×10^{-3} per year. Extrapolated forward, that rate reaches the conjectured floor of 0.2075 only around 2074; reaching it by 2033 would demand roughly seven times the historical rate for the exponent term alone, and even at the floor the work factor remains 2^{182} . The observed pace of algorithmic erosion is therefore about an order of magnitude short of what a near-term break by erosion would require, which is independent, historical support for the structural claim that a 2030s lattice break is contingent on the effective dimension collapse rather than on exponent erosion. The exponents quoted are landmark literature values; the calculation is reproducible from the released materials (Appendix A).

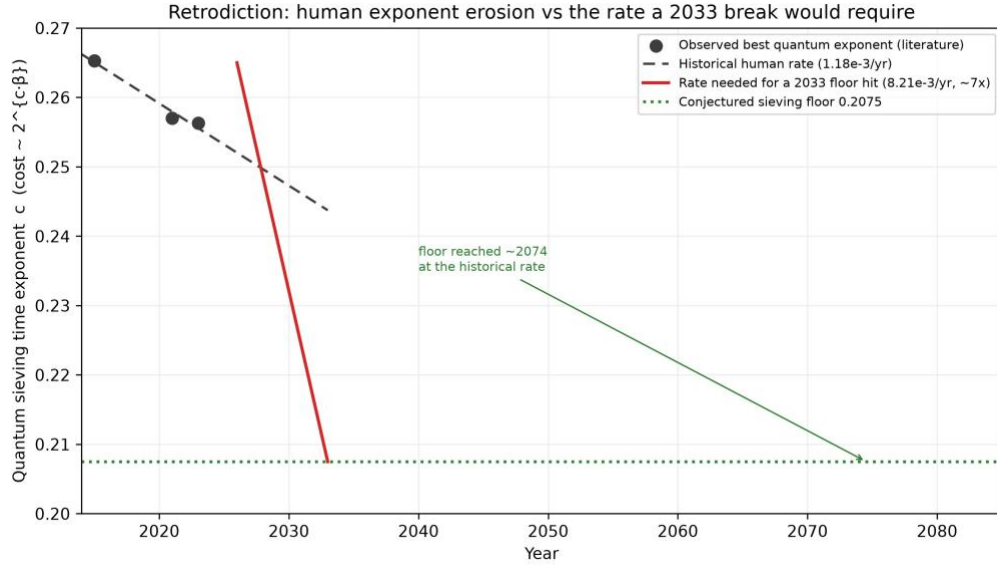


Figure 8C: Retrodiction of the quantum sieving time-exponent trend: observed human progress (about 1.2×10^{-3} per year) versus the rate a 2033 floor crossing would require (about seven times faster). Landmark literature values; reproducible from the released materials (Appendix A).

Worked Examples in Appendix C, ML-DSA-44 (FIPS 204) and SLH-DSA (FIPS 205). Applying the same condition to ML-DSA-44, whose security rests on Module-LWE and Module-SIS at a smaller block size (β on the order of 420), with a slightly larger budget $B(2036) \approx 92$ bits, gives $(1 - \rho) \leq 92 / (0.265 \cdot 420) \approx 0.83$, that is $\rho \geq 0.17$ for a 2036 crossover: a smaller required collapse than Kyber, consistent with the later date and larger budget. SLH-DSA is structurally different. As a stateless hash-based signature it contains no lattice for ρ to act on; its only quantum exposure is Grover-type preimage and collision search, a quadratic effect already absorbed into its conservative parameters, with a classical work factor on the order of 2^{128} at the lowest security level. An idealized Grover search would need only about 2^{64} quantum evaluations, but the search is essentially sequential, and under the realistic circuit-depth limits that NIST applies (the same accounting given for AES-256 in Appendix C) the effective cost remains near the classical value, a level the hardware budget $B(t)$ reaches only at approximately 2045, at the very edge of the projection horizon. The modeled SLH-DSA crossover (approximately 2036 under Track 3 and 2041 under Track 2 in Appendix C) arrives well before that and, like the AES-256 curve, is not produced by any concrete mechanism and rests on the same hypothesized novel-attack assumption flagged for AES in Appendix C. Applying the cost model therefore separates the figures into three honesty tiers: mechanism-backed (RSA and ECC, via Shor), contingency-backed (Kyber-768/1024 and MLDSA-44, via the explicit and falsifiable ρ), and hypothesis-only (SLH-DSA and AES-256).

Assumptions and limitations. This Track 0 result is itself a guard against alarmism: the model's own no-AGI baseline places every lattice and hash break beyond the horizon, so the near-term lattice crossovers are properties of the AGI-gated software driver and are presented as conditional scenarios, not predictions.

Memory-bounded break condition. The break condition above prices time (operation count) only. Lattice sieving is also memory-bound: the sieve list holds $\approx 2^{0.2075 \cdot d}$ vectors, 2^{182} for MLKEM-1024 ($\beta = 877$) and $2^{129.7}$ for ML-KEM-768, and a budget expressed as the base-2 logarithm of affordable operations does not purchase that memory. The 2^{182} that appears above as a time work-factor floor is therefore also the

memory requirement; a structural collapse acting on the block size β must collapse the memory requirement as well, so the stated collapse magnitudes (for example $\rho \geq 0.66$ for ML-KEM-1024) understate what the hypothesis must achieve. Adding the memory constraint pushes the contingency-tier crossovers later and strengthens the paper’s claim not to be over-forecasting.

Both sieving exponents as a band. Because the quantum sieving exponent 0.265 assumes idealized QRAM (Section 3.1.3 notes that the classical 0.292 may be closer to the operative exponent once realistic memory-access and circuit-depth costs are charged), the required collapse fractions are reported as a band. At 0.292 they rise to 0.5616 (ML-KEM-768), 0.6876 (ML-KEM-1024) and 0.2498 (ML-DSA-44), against 0.5170, 0.6558 and 0.1734 at 0.265; the ML-DSA-44 figure moves roughly 44% in relative terms, which is not a rounding matter. Each fraction follows from $\rho = 1 - B/(c \cdot \beta)$ with block sizes $\beta = 625/877/420$ and budgets $B = 80$ (ML-KEM, 2033) and 92 (ML-DSA-44, 2036).

Cost-model boundary. The cost model does not price several effects: dual attacks (cited but not incorporated), multi-target and amortized cost across many keys, decapsulation-failure attacks, the cost of discovering the geometric embedding as distinct from exploiting it, and classical cryptanalytic progress before AGI onset. Pre-AGI classical progress is folded into the AGI-gated driver at 0.04 per year; because human lattice reduction has improved steadily for two decades, attributing almost all non-Shor progress to AGI overstates that parameter’s explanatory weight.

4.6 Progress Timeframes for CRQCs and AI

Given how quickly both quantum computing and AI have advanced in the last few years, the biggest opportunities for potential improvements in quantum algorithms or quantum computing technologies are likely to be driven by CRQC+AI. This section presents these Q-Day Spectrum threats to the different algorithms in a single chart and briefly highlights some of the potential improvements related to CRQC+AI.

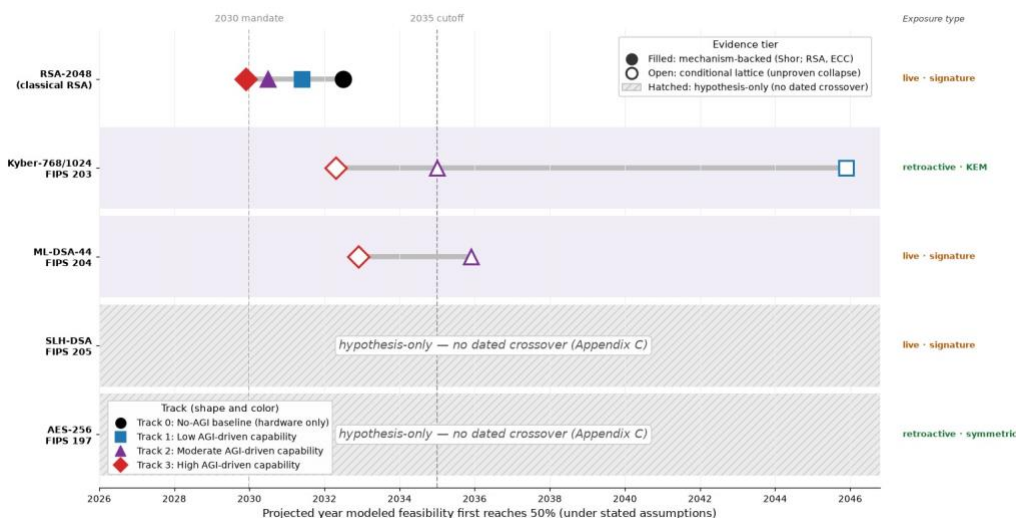


Figure 9: CRQC+AI Vulnerability Spectrum for TLS 1.3: Projected Year Modeled Feasibility

Crossover First Reaches 50% by Algorithm and Track; hash-based and symmetric primitives are shown as hypothesis-only bands without dated crossovers

Note: the lattice, hash-based, and AES-256 crossovers shown are conditional stress scenarios under the stated model assumptions, not known attacks. Mechanism-backed risk applies only to RSA and ECC via Shor's algorithm. The crossover years are conditional scenario outputs, not forecasts (Section 5). Every Track 2 and Track 3 crossover (and the marginal end-of-horizon Kyber-1024 Track 1 crossover) is gated on the assumed 2028 AGI onset: a two-year slip in that onset moves those crossovers later by approximately the same amount.

Model-derived progress timeline: Figure 9 consolidates the crossover estimates from Figures 6A/6B through 8A/8B and the Appendix C stress-case Figures C1A/C1B and C2A/C2B into a single view that defines the spectrum of risk the following subsections describe. The combined hardware-and-software model places the earliest feasible breaks against classical RSA-2048 at approximately 2030 under the most aggressive Track 3 assumptions, with even the no-AGI Track 0 baseline crossing by roughly 2032 on hardware scaling alone. The lattice-based PQC schemes are essentially unaffected by Track 0 and Track 1 (neither hardware scaling nor low software-driven capability threatens them; the sole exception is a marginal Kyber-1024 Track 1 crossover at 2045.9, at the very edge of the horizon) and first become vulnerable in practice under the AGI-software tracks: Kyber-768/1024 (FIPS 203) at approximately 2032 (Track 3) to 2035 (Track 2), and ML-DSA-44 (FIPS 204) at approximately 2033 (Track 3) to 2036 (Track 2). The two parameter sets share a single plotted crossover only because the capability model assigns them equal couplings, not because their hardness is equal, since the cost model of Section 4.5 shows the Kyber-768 hybrid requires a smaller collapse ($\rho \geq 0.52$) than standalone Kyber-1024 ($\rho \geq 0.66$). The hash-based SLH-DSA (FIPS 205) and symmetric AES-256 (FIPS 197) are shown in Figure 9 only as hypothesis-only bands without dated crossovers, for the reasons given in Appendix C; both are more durable than the lattice schemes, with AES-256 the most resistant primitive evaluated. Their model-generated dates are reported, with the accompanying no-cryptanalytic-basis caveat, only in Appendix C. The phase windows below are framed to follow this model-derived sequence. These phases describe the conditional scenario sequence the model produces under its stated assumptions, not predicted events; the lattice and hash phases are gated on the unproven ρ collapse and inherit the honesty tiers of Section 4.5. They are also sensitive to the assumed 2028 AGI onset: because the lattice, hash-based, and symmetric schemes depend almost entirely on the AGI-gated software driver, a two-year slip in that onset moves each of their Track 2 and Track 3 crossovers later by roughly one and a half to two years, while the hardware-driven RSA Track 0 crossover is unchanged.

KEM and signature exposure are not commensurable. A KEM break is *retroactive* against harvested traffic and drives the HNDL argument, governed by Mosca's secrecy horizon X; a signature break requires a *live* attacker against a certificate still within validity, governed by certificate and trust-anchor lifetime. The two therefore do not share a single feasibility axis: ML-DSA-44 represents live exposure and ML-KEM represents retroactive exposure.

4.6.1 Hardware-Driven CRQC Scaling and Early RSA/ECC Exposure

Hybrid Quantum-Classical Algorithms: Quantum computing capabilities advance significantly with mainstream adoption of hybrid quantum-classical frameworks.

Quantum-Enhanced AI Optimization: CRQC+AI begins leveraging quantum-enhanced optimization algorithms like quantum annealing and VQE, improving efficiency in optimization and linear algebra applications.

Initial Cryptanalysis Insights: Early academic research explores the combined potential of the HHL algorithm and low-dimensional representations in geometric algebra, probing for, but not yet establishing, theoretical vulnerabilities in lattice-based PQC schemes.

4.6.2 RSA/ECC Collapse and Possible First Lattice (Track 3) Breaks

Quantum Algorithm Progress: Quantum hardware reliably demonstrates implementation of advanced algorithms, including HHL-type linear-algebra subroutines at small scale; as Section 2.1 establishes, however, such demonstrations do not by themselves yield a practical exponential advantage against the dense, ill-conditioned, cryptographic-size lattice problems underlying LWE.

Quantum-Geometric Proof-of-Concept Attacks: In scenarios where the conjectured geometric embedding is found, researchers would attempt limited-scale proof-of-concept attacks on lattice-based PQC schemes by combining quantum-accelerated optimization with geometric algebra; whether such attempts confirm any practical vulnerability is the open question identified in Sections 2.2 and 4.5, not a settled result.

Industry Reassessment of PQC: The cybersecurity industry reassesses migration timelines, treating current lattice-based implementations in TLS 1.3 as potentially at risk should such weaknesses be confirmed.

4.6.3 Potential Lattice PQC Compromise (Kyber-768/1024 and ML-DSA-44) Quantum AI Scaling: Significant quantum hardware advancements enable practical hybrid quantum-AI algorithms at intermediate scales (1000 to 5000 qubits), a precondition for, though not a guarantee of, viable cryptanalysis of lattice-based encryption schemes.

Cryptanalytic Breakthroughs: This phase is reached only if a geometric embedding with a nonzero collapse fraction ρ is discovered; in that contingency, quantum-enhanced geometric methods would reduce the effective hardness of lattice problems enough to enable attacks on previously secure PQC schemes. Absent such a mechanism, the phase does not occur.

TLS 1.3 PQC Exposure: Should the preceding contingency hold, industry would treat PQCbased TLS 1.3 deployments as exposed and anticipate real-world vulnerabilities on a short horizon; absent that contingency, this exposure does not arise. The joint Monte Carlo (Table 7, Appendix A.5) reinforces that this is a contingency rather than a hardware timeline: under hardware-only Track 0, ML-KEM-1024 shows a median crossover of 2048.8 and effectively zero probability of exposure by 2035.

4.6.4 Possible Hash-Based Signature Exposure and First Symmetric-Layer Threat

Advanced CRQC+AI Algorithms: Robust CRQC+AI algorithms, leveraging both the enhanced capabilities of AI-enhanced quantum algorithms such as HHL and LCHS with geometrically simplified cryptographic representations, would become practical and scalable only if the geometric simplification is realized, on which condition they could facilitate broader cryptanalysis.

Real-World Cryptographic Compromise: Should the preceding contingency materialize, quantum-geometric attacks would expose vulnerabilities in deployed lattice-based PQC schemes, including within TLS 1.3.

Urgent Cryptographic Standard Revisions: Should such compromises occur, NIST and global standardization bodies would expedite the revision and recommendation of quantum-AI-resistant cryptographic methods, prompting swift industry adoption.

4.6.5 Residual Symmetric-Layer Exposure and Post-AGI Outlook

Accelerated Cryptographic Evolution: The arrival of AGI coupled with recognition of CRQC+AI threats should lead to rapid, industry-wide transitions to dynamically agile and hybrid cryptographic practices.

Continuous Quantum Resilience: Organizations will need to embrace dynamic cryptographic agility, leveraging quantum-AI-driven methods and geometric algebra insights to maintain long-term quantum resilience.

To place these scenario outputs alongside external expectations, Table 4 compares the model’s crossover ranges with the principal published quantum-threat timelines and migration mandates. The model is not a forecast and does not claim greater precision than the expert surveys; the comparison is included so the scenario range can be read against the policy deadlines and hardware roadmaps that motivate it. The table also situates the model in its methodological family: like the resource-estimation and expert-elicitation entries beside it, it is intended to be revised as evidence accumulates, differing chiefly in that its revision rule is stated explicitly in Section 3.3.5.

Source	Position and what it provides	Indicative timing
Mosca / Global Risk Institute [50],[80], [69]	Expert-survey risk estimate (Mosca inequality framing)	One in seven by 2026, about 50% by 2031; 2025 survey median CRQC 2029 to 2032 (34% by 2030)
NIST IR 8547 [51]	Standards and migration policy	RSA-2048 and ECC P-256 deprecated by 2030, disallowed by 2035
NSA CNSA 2.0 [164]	National-security migration mandate	Quantum-resistant support by 2025; category deadlines 2030 to 2033; full NSS migration by 2035
IBM roadmap [101]	Quantum hardware roadmap	First large-scale fault-tolerant system (Starling, about 200 logical qubits) by 2029; cryptanalytically relevant scale later
Google Quantum AI / Gidney (2025) [82]	Resource estimate and hardware roadmap	RSA-2048 factoring resource cut to under one million physical qubits (from about 20 million); error-corrected machines targeted in the early 2030s
This paper (capability model)	Scenario stress test, not a forecast	RSA mechanism-backed crossover about 2030 to 2032; conditional lattice, hash-based, and symmetric scenarios 2032 to 2046

Table 4: Comparison of the model’s crossover ranges with principal published quantum-threat timelines, hardware roadmaps, and migration mandates.

5. Limitations and Falsifiability

This section consolidates the study’s limitations and states the conditions under which its conclusions would fail, so the model can be judged as a falsifiable instrument rather than an open-ended projection.

Limitations. The capability model that generates the figures uses scenario couplings, the hardware sensitivities α and the software sensitivities β , that are chosen rather than measured; they encode judgments about how strongly each primitive is eroded by quantum hardware and by AI-assisted attacks, and different analysts could reasonably choose different values. The crossover years are therefore outputs

of those chosen inputs, not empirical predictions, and the reproducibility documented in Appendix A means the scenario can be regenerated rather than that the dates are forecast. The runtime axis of the figures is likewise a normalized index anchored at a common baseline for all five schemes, not a per-scheme physical runtime estimate (Appendix A.1). The model holds the quantum algorithm fixed and does not attempt to discover new cryptanalysis; it asks only how much engineering and AI-assisted progress would be required for earlier compromise. The 2028 AGI onset is an assumption drawn from interested-party roadmaps, and the sensitivity analysis in Appendix A.4 shows it is the single most consequential input for every software-driven crossover. The three evidence tiers are treated as qualitatively distinct, and no claim is made that the lattice, hash-based, or symmetric crossovers correspond to any known attack.

How this model could be wrong. The central conclusions do not hold, and the corresponding curves should be disregarded, under any of the following conditions.

- If AI produces no durable algorithmic improvement to cryptanalysis, so that the software couplings β remain effectively zero, the lattice, hash-based, and symmetric tracks never cross within the horizon and only the mechanism-backed RSA and ECC exposure remains.
- If no effective-dimension-collapse or comparable structural advance against lattices ever materializes, so that ρ stays at zero in the Section 4.5 cost model, the contingency tier is void. If Harrow-Hassidim-Lloyd and related quantum linear-algebra methods remain inapplicable to the noisy bounded-distance-decoding structure of LWE, a central hypothesized mechanism is removed.
- If quantum hardware scaling stalls, with the hardware rate r_{hw} approaching one so that the hardware driver flattens, even the RSA crossover moves out by years, as the sensitivity analysis shows. If fault-tolerant error correction plateaus below cryptanalytically relevant scales, the mechanism-backed timeline lengthens regardless of algorithmic progress.
- And if AGI arrives substantially later than the assumed 2028, every software-driven crossover slips by approximately the same amount.

Each of these is an observable, falsifiable condition rather than a matter of interpretation. The channel-level tracking indicators that operationalize these conditions, hardware milestones, error rate trends, decoder latency, and resource-estimate movement, are given in Section 3.3.5; each indicator updates a specific acceleration factor, rather than the projection in its entirety. Read together, the falsification conditions above and the recalibration protocol of Section 3.3.5 make this a living estimation instrument: it is expected to err in measurable ways and measuring that error against milestones as they become historical data points is the mechanism by which successive, versioned reruns of the model improve its tracking of the record and potentially its projections of future changes in vulnerability spectrums.

6. Implications for Migration and Governance

The modeling above measures primitive erosion in TLS 1.3. Three consequences for migration and governance follow, none of which depends on the effective-dimension-collapse conjecture; each is a present requirement rather than a projection.

6.1 Crypto-Agility as a Control Objective

Crypto-agility is not a technical property of a stack; it is a governance capability, comprising cryptographic inventory, asset ownership, rotation authority, change-control process, exception handling, and evidence retention. [174] A deployment is agile only if it can enumerate every primitive in use, name the owner

authorized to rotate it, execute a rotation under change control, and retain evidence of the change. Recommending crypto-agility without these leaves the reader with a purchase decision rather than a control objective. The migration decision, and the authority under which it is exercised, belongs to the cryptographic-inventory owner, not to the protocol.

6.2 Sector Data-Lifetime Spectrum (Mosca’s X and Y)

Mosca’s inequality $X + Y > Z$ is a sector-specific decision rule. This paper models Z, the capability horizon, three times, but never populates X, the secrecy horizon, or Y, the migration duration.

Representative values are shown Table 5.

Sector	Secrecy horizon X	Migration duration Y	Urgency vs modeled Z
Payments / card data	~5–10 yr [178]	months–2 yr (cloud)	lower
Health records	~25–50+ yr [177]	2–5 yr	high
Legal / privileged	decades [176]	2–5 yr	high
National security	~25–75+ yr [175]	5–10+ yr (OT/embedded) [179]	highest

Table 5: Representative sector data-lifetime spectrum (Mosca’s X and Y). Secrecy horizons X and migration durations Y are sourced in the running text.

The secrecy horizons X are anchored in published mandates: the 25-, 50-, and 75-year declassification tiers for national-security information [175]; the survival of attorney-client privilege beyond the client’s death [176]; HIPAA documentation retention and state medical-record statutes together with the patient-lifetime confidentiality of health data, so the operative horizon is the patient’s life rather than the shorter retention minimum [177]; and SEC Rule 17a-4 and PCI DSS retention for payment-card data [178]. The migration durations Y are engineering estimates reflecting operational-technology and embedded-system lifecycles of 10 to 20-plus years and published enterprise PQC-migration analyses [179]. The point stands regardless: migration urgency varies by sector, which cuts against a single uniform timeline.

6.3 Sovereignty and Jurisdiction

Crypto-agility is a present requirement on legal grounds alone, demonstrable today and requiring no speculative cryptanalysis. Algorithm mandates already diverge: CNSA 2.0 specifies standalone ML-KEM-1024 for United States national-security systems, against a widely deployed hybrid default (ECDHE with ML-KEM) and differing European and Australian positions. Trust-anchor jurisdiction, HSM sourcing, and cross-border key custody differ by sovereign. An organization operating across jurisdictions must be able to switch algorithms, trust anchors and key custody on legal timelines it does not control, which is precisely a crypto-agility capability. This is a stronger argument for the paper’s own conclusion than the effective-dimension-collapse hypothesis, and it depends on no unproven mechanism.

6.4 Issuers, Trust Anchors and Certificate Lifecycle

Post-quantum migration in TLS 1.3 is not only a matter of the negotiated primitives; it depends on who issues certificates and how trust anchors are rotated. Root and intermediate certificate authority keys, and the trust anchors embedded in operating systems, browsers and devices, have lifetimes measured in years to decades and are updated on distribution cycles the relying party does not control. A post-quantum chain requires post-quantum-capable issuers, post-quantum-aware path validation, and a trust-anchor distribution that reaches long-lived and embedded clients, and it typically lags the negotiated key

exchange. None of this requires a cryptographically relevant quantum computer to become a live constraint; it is an ecosystem exposure that materializes on today's issuance and distribution timelines. It lies outside the primitive-erosion instrument of this paper and is flagged here as a dependency a TLS 1.3 migration program must plan for separately.

6.5 Interoperability and Negotiation

TLS 1.3 negotiates its algorithms, which is a strength, but negotiation is also where a migration succeeds or fails in practice. A hybrid key exchange, for example the concatenation of ML-KEM with an elliptic-curve exchange in draft-ietf-tls-ecdhe-mlkem, [161] must be offered, understood and preferred by both endpoints; middleboxes, session-resumption caches and libraries that do not recognize the new groups can cause negotiation failure or a silent downgrade to a classical exchange. Interoperability across a heterogeneous estate, and resistance to downgrade, are engineering problems that arrive with deployment rather than with a quantum computer. This paper models the primitives on the assumption that a post-quantum or hybrid group is negotiated end to end; the negotiation and interoperability layer that makes that assumption hold is out of scope and is noted here as a distinct control.

6.6 Entropy and Key Provenance

Every guarantee in this paper is conditioned on keys drawn from a sound entropy source. A postquantum primitive keyed from a weak or predictable generator is no stronger than its randomness, independent of any quantum advance. Entropy quality, seeding on constrained and embedded devices, and the provenance of keys, where they were generated, by what module, and under what assurance, are recurring sources of real-world failure that require no cryptographically relevant quantum computer to exploit. Key provenance also bears on HNDL and harvest-now, forge-later exposure: keys generated or held outside the operator's control widen the surface for retroactive decryption. Entropy and key provenance are properties of the implementation and the operating environment rather than of the negotiated primitive, so they lie outside the instrument used here and are recorded as a further ecosystem dependency.

7. Conclusion

Depending upon the timeline at which CRQC+AI progresses, the potential for large-scale financial loss for TLS 1.3 could begin as early as 2030 with attacks against RSA. The compromise of PQC encryption methods would have similar effects to a compromise in any use of RSA encryption and the risk vulnerability spectrum could begin as early as 2032. These two dates carry different evidential weight: the RSA and ECC exposure is mechanism-backed, resting on Shor's algorithm, whereas the PQC dates are conditional scenario outputs that depend on a structural advance against lattices that has not been demonstrated. However, the methods utilizing AGI+CRQC+AI as outlined herein would magnify these issues because of the rapidity and generalization ability of such attacks. Effects would include the following:

Table 6 maps each class of cryptographic break to the primitive or key material compromised and its effect on TLS 1.3.

Threat class	Example	TLS 1.3 effect
Algorithmic public-key break	Shor against RSA or ECC	Certificate forgery, identity compromise, legacy decryption
KEM or key-exchange break	ML-KEM structural break	Session-key recovery and harvest now-decrypt-later exposure

Signature break	ML-DSA or SLH-DSA break	Authentication failure, impersonation
Symmetric cipher break	AES structural break	Bulk confidentiality failure
Implementation or side-channel break	Timing, fault, or cache leakage	Key theft without breaking the primitive

Table 6: Threat classes and their TLS 1.3 effects.

Eavesdropping on Communications: By intercepting PQC key exchanges, an attacker could extract the shared keys between parties, allowing them to decrypt sensitive communications.

Man-in-the-Middle Attack: An attacker could position themselves between two communicating parties on a network, intercepting and altering messages once the keys used for encryption are obtained and hacked, thus compromising the integrity of the communication.

Data Theft: With access to encryption keys, an attacker could decrypt stored data on servers or devices, leading to unauthorized access to sensitive information such as personal data, financial records, trade secrets.

Impersonation: By obtaining cryptographic keys, an attacker could impersonate a legitimate user or service, allowing them to perform unauthorized actions or transactions under false pretenses.

Disruption of Services: An attacker could use the decrypted private keys to disrupt services by altering or injecting malicious data into communications, potentially leading to denial-of-service conditions or other operational failures.

Access to Encrypted Backups: If an attacker gains access to private encryption keys, they could decrypt backups of sensitive data, expose historical information and alter data that may not be actively monitored.

Exploitation of IoT Devices: Many Internet of Things (IoT) devices rely on private cryptographic keys for secure communication. An attacker could exploit these keys to gain control over IoT devices, leading to privacy breaches or physical security risks.

Manipulation of Smart Contracts: In blockchain environments, if an attacker obtains the private keys used to sign transactions, they could manipulate smart contracts, leading to financial losses or fraudulent activities.

The potential for massive and rapid financial loss because of CRQC+AI vulnerabilities is substantial and difficult to bound precisely. Just as it took decades before the British and Americans revealed they had cracked the Enigma cryptography machine in World War II, we won't know when hackers will be effective at using CRQC+AI resources to break PQC encryption algorithms.

When weighed against the relatively low cost and high probability of successful defenses to PQC vulnerabilities through research and use of crypto-agile hybrid cryptographic approaches that are now being advocated for by NIST, [9] the case is compelling for acting now to mitigate these risks not by assuming that PQC will be the last encryption migration needed. Instead, as some business executives are beginning to recognize, the need for migration to PQC is better viewed as a trigger for needing to implement crypto-agile approaches. [165] Layered risk scoring models for TLS connections under quantum threats can further help organizations identify and prioritize their most exposed connections during this transition. [166] Beyond these practical conclusions, the durable contribution of this paper is the estimation instrument itself: a reproducible, parameterized scenario model whose parameters can be varied, rerun, and recalibrated against emerging milestones through the update mechanics of Section

3.3.5, complementing quantum resource-estimation studies and expert-elicitation timeline surveys with an auditable procedure for converting future observations into revised risk timelines.

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Data Availability Statement: No new empirical data were created or analyzed in this study. The figures are generated from the parameterized model described in Sections 3 and 4, whose parameters and assumptions are disclosed in the text.

Conflicts of Interest: N.G. and B.P. are employees of EnQuanta (Voicelt Technologies, Inc dba EnQuanta), and E.U and M.H. are consultants of EnQuanta, a company that develops commercial post-quantum and crypto-agile cryptographic products and services; accordingly, they have a financial interest related to the crypto-agile and hybrid cryptographic approaches discussed in this work.

Appendix A. Reproducibility of the Combined Hardware and Software Capability Model

This appendix documents the computational model used to generate the runtime and feasibility figures (Figures 6A/6B through 8A/8B, Figure 9, and Appendix C Figures C1A/C1B and C2A/C2B), so that every curve can be reproduced from explicit equations and parameters. The figures are produced by a single combined hardware and software capability model, set out below, which is the generative instrument behind the curves. It is complementary to the analytical framings in the body: the channel model of Section 3.3 and the cost model of Section 4.5 are separate instruments that bound the same question, while the capability model here is the one that produces the plotted curves. The model holds the quantum algorithm fixed: AI accelerates engineering and the discovery and optimization of attacks, but it does not change the polynomial scaling of Shor's algorithm, lower the fault-tolerant error-correction threshold, or repeal coherence limits.

Appendix A.1. Governing Equations

Cryptanalytic capability $C(a, k, t)$, for algorithm a on track k in calendar year t , is measured as the number of base-10 orders of magnitude (in seconds) by which the estimated attack runtime has been reduced from its starting value. It is the sum of a hardware term and a software term:

$$C(a, k, t) = \alpha(a) \cdot D_{hw}(t) + \beta(a, k) \cdot D_{sw}(k, t)$$

The hardware driver D_{hw} is an accelerating (convex) accumulation of effective quantum resources, present in every track including Track 0, because engineering progress is not gated by AGI:

$$D_{hw}(t) = A \cdot (r_{hw}^{(t - t_0)} - 1) / (r_{hw} - 1), A = 0.34, r_{hw} = 1.20, t_0 = 2026$$

The software driver D_{sw} is gated on the 2028 AGI onset and grows at a track-specific rate, with a small pre-AGI human-progress term:

$$D_{sw}(k, t) = 0.04 \cdot (t - t_0) + B_k \cdot (r_k^{(t - t_{AGI})} - 1) / (r_k - 1), t_{AGI} = 2028$$

In both drivers the exponent is clamped at zero below its onset year, so the hardware term contributes only from t_0 and the AGI term only from t_{AGI} . The coupling $\alpha(a)$ is the sensitivity of algorithm a to raw hardware (Shor), large for RSA and near zero for the lattice, hash, and symmetric schemes; $\beta(a, k)$ is its sensitivity to the software attacks enabled on track k . The estimated attack runtime is $R(a, k, t) = R_0 - C(a, k, t)$ on a base-10 log-seconds axis, with $R_0 = 10$ (Centuries), a common visual baseline shared by all five schemes rather than a per-scheme empirical runtime estimate. $R(a, k, t)$ is therefore a normalized runtime index: within a scheme it tracks modeled progress over time, but cross-scheme differences on this axis carry no information about relative attack cost, which is treated in the cryptanalytic units of Section 4.5. Modeled feasibility is a logistic of the margin by which runtime has fallen below the one-month line:

$$P(a, k, t) = 100 / (1 + \exp(-(MON - R(a, k, t)) / 0.85)), MON = \log_{10}(2.6 \times 10^6) \approx 6.415$$

A scheme reaches its 50 percent modeled-feasibility crossover in the first year for which the runtime falls to the one-month line, equivalently when $C(a, k, t)$ reaches $R_0 - MON \approx 3.585$. No crossover year is taken from prior or external figures; every crossover emerges from these equations.

Appendix A.2. Parameters and Their Sources

Global constants: the reference year $t_0 = 2026$; the AGI onset $t_{AGI} = 2028$; the runtime axis is base-10 log seconds from Seconds to Centuries, with start value $R_0 = \log_{10}(10^{10}) = 10$ and the one-month feasibility

line at $\text{MON} = \log_{10}(2.6 \times 10^6) \approx 6.415$; the logistic width is 0.85. A scheme crosses 50 percent when capability C reaches $R_0 - \text{MON} \approx 3.585$ orders of magnitude. R_0 is a common visual baseline shared by all schemes for comparability, not a per-scheme estimate of the 2026 attack runtime. On this normalized axis the one-month line MON functions as a fixed capability threshold ($R_0 - \text{MON} \approx 3.585$ orders of magnitude) rather than a claim that any given attack would take one month of physical time.

Hardware driver: $A = 0.34$ and $r_{\text{hw}} = 1.20$, an accelerating accumulation of effective quantum resources of roughly 20 percent per year, present in every track. It is the granular counterpart of the Rose's-Law hardware growth in Section 3.2.2.7 and the channel model in Section 3.3.

Software driver: a pre-AGI human-progress slope of 0.04 per year, plus a track-specific AGI-gated term with rate r_k and ceiling B_k taking the values Track 0 (1.01, 0.00), Track 1 (1.10, 0.16), Track 2 (1.17, 0.26), and Track 3 (1.30, 0.30). The Track 3 rate of 1.30 is what produces its sharply convex collapse.

Per-algorithm couplings, given as α and then β for Tracks 0 to 3: RSA-2048, $\alpha = 0.92$, $\beta = 0, 1.15, 1.75, 2.55$; Kyber-1024, $\alpha = 0.05$, $\beta = 0, 0.06, 0.95, 1.45$; ML-DSA-44, $\alpha = 0.04$, $\beta = 0, 0.05, 0.78, 1.18$; SLH-DSA, $\alpha = 0.03$, $\beta = 0, 0, 0, 0$; AES-256, $\alpha = 0.020$, $\beta = 0, 0, 0, 0$. The α values are hardware (Shor) sensitivities, large for RSA and near zero otherwise; the β values are software-attack sensitivities enabled on each track. These couplings are scenario parameters, not measured constants, chosen so that hardware alone erodes only RSA and the lattice schemes move only once the Track 2 and Track 3 software drivers switch on; the hash-based (SLH-DSA) and symmetric (AES-256) schemes are assigned zero software coupling and report no crossover within the horizon, consistent with the hypothesis-only treatment in Figure 9. The hybrid X25519 + Kyber-768 configuration of Figures 7A/7B is modeled with these same lattice couplings: the hybrid falls only when both components fall, and because the classical X25519 component is mechanism-backed (Shor) and falls years earlier under the hardware track, the lattice component is binding and the hybrid's modeled crossovers coincide with the Kyber values reported below.

This failure ordering is a *result*, not an input. Sweeping the coupling space, a lattice scheme crosses before RSA only under one of two conditions: (a) a hardware coupling above RSA's own 0.92, that is, assuming a lattice scheme more Shor-sensitive than RSA, which is a claim about the algorithm rather than a parameter choice; or (b) on Track 3, a software coupling of about 4.48 for ML-KEM1024 against its published 1.45, roughly triple its own value and $1.76 \times$ RSA's 2.55. Every inversion point lies far outside the published couplings and the $\pm 25\%$ sensitivity ranges of Appendix A.4, so the ordering is structurally forced rather than assumed.

Running the model with these parameters yields the 50 percent crossover years plotted in Figure 9: RSA-2048 at 2032.5, 2031.4, 2030.5, and 2029.9 for Tracks 0 to 3; Kyber-1024 at 2045.9 (Track 1), 2035.0 (Track 2), and 2032.3 (Track 3); ML-DSA-44 at 2035.9 and 2032.9 (Tracks 2 and 3); and SLH-DSA and AES-256, which carry zero software coupling, show no feasible break within the 2046 horizon, as do the remaining tracks.

Appendix A.3. Procedure to Regenerate the Figures (Pseudocode)

CONSTANTS

$t_0 = 2026$; $t_{\text{AGI}} = 2028$

$R_0 = \log_{10}(1e10)$ # normalized index baseline (Centuries), common to all schemes

$\text{MON} = \log_{10}(2.6e6)$ # one-month feasibility line

$A = 0.34$; $r_{\text{hw}} = 1.20$ # hardware driver
 $\text{sw_pre} = 0.04$ # pre-AGI software slope


```

SW = { 0:(1.01,0.00), 1:(1.10,0.16), 2:(1.17,0.26), 3:(1.30,0.30) } # (r_k, B_k) alpha[algo] , beta[algo][track]
# per-algorithm couplings (Appendix A.2)

FUNCTION D_hw(t):
  e = max(t - t0, 0) ; return A * (r_hw**e - 1) / (r_hw - 1)
FUNCTION D_sw(t, track):
  (r, B) = SW[track] ; e = max(t - t_AGI, 0) return sw_pre * max(t - t0, 0) +
  B * (r**e - 1) / (r - 1)
FUNCTION capability(algo, track, t): return alpha[algo]*D_hw(t) + beta[algo][track]*D_sw(t, track)
FUNCTION runtime(algo, track, t): return clamp(R0 - capability(algo,track,t), low = -0.4, high = R0)
FUNCTION feasibility(algo, track, t): margin = MON - runtime(algo,track,t) ; return 100 / (1 + exp(-
margin/0.85))

FUNCTION crossover_year(algo, track): for t from
  2026 to 2046 in 600 steps:
    if feasibility(algo,track,t) reaches 50: return t
  return NONE within horizon

```

Reproduction materials. The runnable counterpart to this appendix, comprising the capability model, the seeded Monte Carlo that regenerates Table 7 and Figure A2, and the figure-generation scripts, is available at <https://enquanta.com/academia/crqc-ai-tls13-reproduction.zip>. The Monte Carlo uses a fixed base seed (20260731) offset by cell index, so each algorithm-and-track cell regenerates independently; with $N = 40,000$ the released scripts reproduce the deterministic crossover years exactly and the Table 7 percentiles to within about 0.2 years. The illustrative curves of Figures 4A and 4B are stylized reconstructions rather than measured series, and the Figure 5 qubit estimates are approximate reads pending the Global Risk Institute source table; the release notes mark each accordingly.

Appendix A.4. Sensitivity Analysis

Because each scheme crosses when its capability reaches about 3.585 orders of magnitude, the sensitivities are clear from the two drivers. For RSA-2048 the capability is dominated by the hardware term ($\alpha = 0.92$), so RSA crosses under every track, hardware alone being sufficient, and the track spread is narrow, about 2.5 years, set by its software couplings; widening that fan requires larger RSA β values. For the lattice, hash, and symmetric schemes α is near zero, so they are essentially immovable under Track 0 (hardware only) and Track 1 (small β), apart from a marginal end-of-horizon Kyber-1024 Track 1 crossover at 2045.9, and they cross materially only once the Track 2 and Track 3 software drivers, gated at 2028, accumulate enough capability; their crossover years are therefore governed by $\beta(a, k)$ and by the track rates r_k and ceilings B_k .

Three inputs move the whole picture. The Track 3 rate $r_k = 1.30$ is the sole source of its sharply convex collapse; lowering it toward the Track 2 value of 1.17 pushes the Track 3 crossovers later and removes the convex curl. The AGI onset $t_{AGI} = 2028$ gates every software-driven crossover, so a later onset delays all lattice, hash, and symmetric breaks while leaving RSA's hardware-driven Track 0 crossover unchanged. The most consequential and most speculative inputs are the AES-256 couplings ($\alpha = 0.020$, $\beta = 0.155$ and 0.210 for Tracks 2 and 3), which are the entire basis for the Track 2 and Track 3 AES breaks; there is no established cryptanalytic mechanism behind them, and setting AES-256 β to zero restores it to an unbroken control.

To make the dependence on each input explicit, the two representative crossovers in Figure A1 were recomputed while varying one parameter at a time across a deliberately widened stress range, roughly double the calibrated ranges used by the Monte Carlo of Appendix A.5. The analysis is deliberately one-at-a-time, isolating each parameter's effect rather than capturing interactions among parameters; a joint treatment over the stated ranges is developed as a Monte Carlo analysis in Appendix A.5. The two panels show a clean separation. The hardware-driven RSA Track 0 crossover (Figure A1, left panel) moves only with the hardware amplitude A , the hardware acceleration rate r_{hw} , and RSA's hardware coupling α , and is completely insensitive to the AGI onset and the software parameters. The software-driven Kyber-1024 Track 3 crossover (Figure A1, right panel) is dominated by the AGI onset year, followed by the software coupling β and the track ceiling B_k , and is nearly insensitive to hardware. Even under these widened ranges, no single one-at-a-time variation shifts either crossover by more than about 3.6 years (the AGI onset in the software-driven case), so no individual parameter choice drives the result on its own; and the failure ordering itself is governed not by any of these one-at-a-time bands but by the coupling inversion threshold of Appendix A.2, which lies far outside even these ranges.

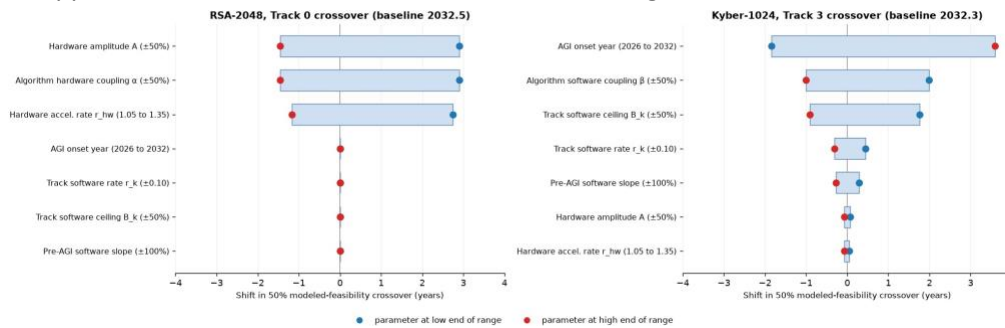


Figure A1. One-at-a-time sensitivity of the 50% modeled-feasibility crossover to model parameters, for a hardware-driven case (RSA-2048, Track 0; left) and a software-driven case (Kyber-1024, Track 3; right).

Each bar spans the crossover shift as the named parameter is varied across its stated range; blue and red mark the low and high ends. The ranges shown are deliberately widened stress ranges, roughly double the calibrated Monte-Carlo ranges of Appendix A.5: $\pm 50\%$ on the couplings, hardware amplitude and track ceiling; ± 0.10 on the track rate; $\pm 100\%$ on the pre-AGI slope; a 1.05 to 1.35 hardware rate; and a 2026 to 2032 AGI onset. Even so, no single parameter shifts either representative crossover by more than about 3.6 years, and the failure ordering remains governed by the coupling inversion threshold of Appendix A.2, which lies far outside even these ranges.

Four named scenarios make the dependence concrete. A two-year slip in the AGI onset, to 2030, leaves the hardware-driven RSA Track 0 crossover unchanged at 2032.5 but moves every software-driven crossover later by roughly 1.8 years (Kyber-1024 Track 3 from 2032.3 to 2034.1; AES-256 Track 3 from 2038.6 to 2040.4). If quantum hardware stalls, with the hardware rate falling to

1.05, the RSA Track 0 crossover slips from 2032.5 to 2035.3 while the software-driven crossovers barely move, since they depend on the AGI-gated software driver rather than on raw hardware. If AI-assisted attack progress is twenty percent weaker, with all software couplings reduced by a fifth, the lattice, hash-based, and symmetric crossovers move later by roughly 0.6 to 1.0 years. And if Track 3 loses its sharply convex character, with its rate dropping to the Track 2 value of 1.17, the most aggressive crossovers retreat, AES-256 Track 3 moving from 2038.6 to 2041.8.

Appendix A.5. Joint Monte Carlo Over the Parameter Ranges

The one-at-a-time analysis of Appendix A.4 isolates each parameter’s effect but does not capture interactions among simultaneously uncertain inputs. This appendix complements it with a joint Monte Carlo that samples all model parameters together over their stated ranges and reports the quantity the framework is ultimately built to produce: $P(T \leq \text{year})$, the probability that a scheme’s feasible-attack crossover arrives on or before a given year. This is precisely the input Mosca’s inequality requires for the capability horizon Z (Section 3.1), for which any nonzero probability mass inside the secrecy horizon X is sufficient to make migration and crypto-agility rational today.

Sampling. For each algorithm and track, $N = 40,000$ independent draws are taken, each sampling uniformly unless noted: the hardware sensitivity $\alpha(a)$ and software sensitivity $\beta(a, k)$ over $\pm 25\%$ of their Appendix A.2 values; the hardware amplitude A over $\pm 25\%$ of 0.34; the hardware rate r_{hw} over $[1.10, 1.25]$; the pre-AGI software slope over $\pm 50\%$ of 0.04; the track rate r_k over ± 0.05 ; and the track ceiling B_k over $\pm 25\%$. For the AGI-gated tracks (1 to 3) the AGI onset t_{AGI} is drawn from a triangular distribution on $[2027, 2028, 2031]$ with 2028 as its mode, reflecting the spread of the developer and expert forecasts surveyed in Section 4; Track 0 carries no software driver and hence no AGI dependence. Each algorithm-and-track cell uses its own independent, deterministically seeded random stream (a fixed base seed offset by the cell index, reported in the reproduction script), so every result regenerates exactly and independently of the order in which the cells are computed; crossovers beyond 2060 are treated as censored.

Results. Figure A2 plots the resulting $P(T \leq \text{year})$ curves and Table 7 summarizes their 10th-percentile, median, and 90th-percentile crossover years with selected exceedance probabilities. The mechanism-backed RSA-2048 exposure is robust to the AGI question: even the no-AGI Track 0 baseline has a median crossover near 2033 and about a 95% probability of a feasible attack by 2035. The lattice schemes remain outside the policy-relevant window under Track 0 and Track 1 (median crossovers in the late 2040s) and enter it only under the AGI-gated software tracks: Kyber-1024 spanning roughly 2032 to 2035 under Track 3 ($P(\leq 2033) \approx 0.52$, $P(\leq 2035) \approx 0.98$) and roughly 2034 to 2038 under Track 2 ($P(\leq 2035) \approx 0.28$), with ML-DSA-44 about a year later. The joint spread is modestly wider than the one-at-a-time bands of Appendix A.4, as expected once parameters move together, but it does not alter the qualitative ordering or shift any median by more than about a year from the point estimates of Appendix A.2, and the AGI onset remains the dominant source of variance for every software-driven crossover.

Scheme	Track	P10	Median	P90	$P(\leq 2035)$
RSA-2048	0 (no AGI)	2031.8	2033.0	2034.5	0.95
RSA-2048	3	2029.8	2030.5	2031.2	1.00
Kyber-1024	0 (no AGI)	2044.5	2048.8	2055.8	0.00
Kyber-1024	2	2034.5	2035.8	2037.5	0.28
Kyber-1024	3	2032.0	2033.0	2034.5	0.98
ML-DSA-44	3	2032.5	2033.8	2035.0	0.91

Table 7: Joint Monte Carlo crossover distribution (10th percentile / median / 90th percentile, in calendar years) and probability of a feasible attack by 2035, over the parameter ranges of Appendix A.5 ($N = 40,000$ per row).

Interpretation. The Monte Carlo adds no evidence; it propagates the stated uncertainty in the chosen couplings into an explicit distribution, so the crossover years can be read as probability curves rather than single dates. It inherits every assumption and limitation of the capability model, including that the couplings are chosen rather than measured (Section 5); widening or shifting the input ranges shifts the output curves correspondingly. The hash-based and symmetric primitives are deliberately omitted from

these probability statements, for the same reason their dates are withheld from Figure 9: their crossovers rest on a hypothesized mechanism with no cryptanalytic basis (Appendix C), and a probability attached to them would misrepresent a modeling assumption as a calibrated likelihood.

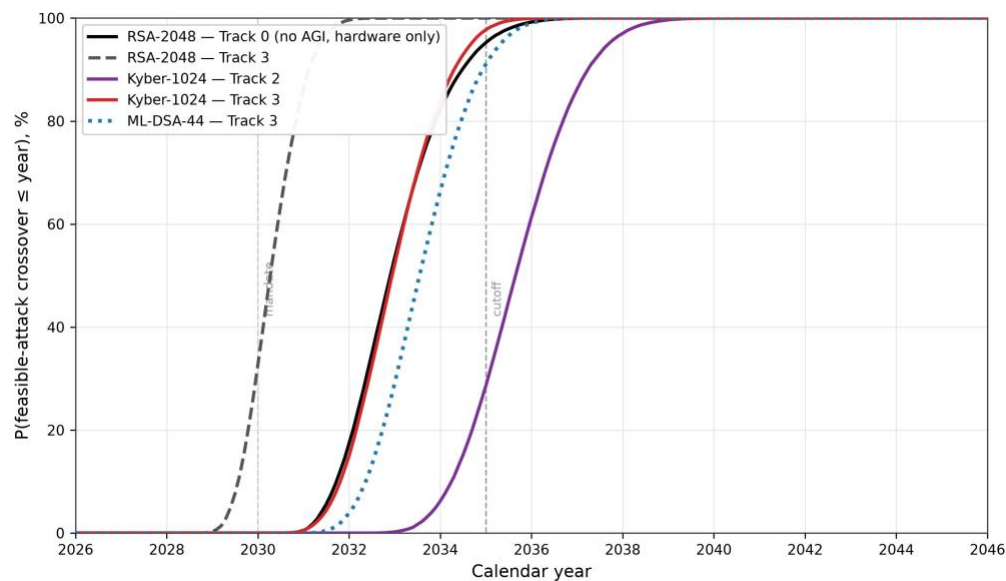


Figure A2. Joint Monte Carlo estimate of $P(T \leq \text{year})$, the probability that the modeled feasible-attack crossover occurs on or before a given year, for the decision-relevant algorithm and track combinations, over the parameter ranges of Appendix A.5.

Appendix B. Notation and Symbols

The model spans the acceleration-factor framing of Section 3.3, the cost model of Section 4.5, and the capability model of Appendix A. Table 8 lists the symbols used across these and disambiguates the two distinct uses of β .

Symbol	Meaning	Symbol	Meaning
t	Calendar year (model time variable)	$R(a, k, t)$	Estimated attack runtime, $R_0 - C$, on a base-10 log-seconds axis
t_0	Reference year, 2026	R_0	Runtime baseline, $\log_{10}(10^{10}) = 10$ (Centuries); a shared visual baseline
t_{AGI}	Assumed AGI onset year, 2028	MON	One-month feasibility line, $\log_{10}(2.6 \times 10^6) \approx 6.415$
$C(a, k, t)$	Cryptanalytic capability for algorithm a on track k , in base-10 orders of magnitude of attack-runtime reduction	$P(a, k, t)$	Modeled feasibility; logistic of the margin by which runtime falls below the one-month line
$D_{hw}(t)$	Hardware capability driver; accelerating, present on all tracks	A_x	Engineering-channel acceleration factor (Section 3.3)
$D_{sw}(k, t)$	Software capability driver; AGI-gated and track-specific	τ_x	Per-channel time to readiness (Section 3.3)
A	Hardware driver amplitude (0.34)	Δ	Sieving-exponent erosion (Section 4.5 cost model); the eroded quantum sieving exponent is $0.265 - \Delta$, floored so that $0.265 - \Delta \geq 0.2075$

r_{hw}	Hardware acceleration rate (1.20)	Δ_{max}	Ceiling of the erosion schedule; $\Delta(t) = \Delta_{max} \cdot (1 - e^{(-R \cdot (t - t_{AGI}))})$ for $t \geq t_{AGI}$, and $\Delta(t) = 0$ before t_{AGI} (Section 4.5)
sw_{pre}	Pre-AGI software progress slope (0.04 per year)	R	Research-acceleration factor in the erosion schedule: human-cryptanalyst-years compressed into one calendar year (Section 4.5); distinct from the runtime $R(a, k, t)$ above
r_k	Track k software acceleration rate (Tracks 0 to 3: 1.01, 1.10, 1.17, 1.30)	$B(t), B_0$	Feasible attack budget in bits, $B(t) = B_0 + r \cdot (t - t_0)$, with slope r (Section 4.5); distinct from the track ceiling B_k above
B_k	Track k software ceiling (Tracks 0 to 3: 0.00, 0.16, 0.26, 0.30)	ρ	Effective block-size collapse fraction (Section 4.5); $\rho = 0$ means no structural advance
$\alpha(a)$	Algorithm sensitivity to hardware (Shor); large for RSA, near zero otherwise	$\delta(\beta)$	Root-Hermite factor of the lattice estimate (Section 3.1.3); governs achievable basis quality at block size β
$\beta(a, k)$	Algorithm sensitivity to the software attacks enabled on track k (capability model)	β (lattice)	Core-SVP block size for the lattice estimate (Sections 3.1.3 and 4.5); distinct from the capability-model coupling $\beta(a, k)$ above

Table 8: Notation and symbols used in the model.

Appendix C. Hypothesis-Only Stress Cases: SLH-DSA (FIPS 205) and AES-256 (FIPS 197)

This appendix presents the two hypothesis-only stress cases separately from the mechanism-backed and contingency-backed projections of Section 4. No known quantum, classical, or algorithmic attack threatens SLH-DSA or AES-256 within the modeled horizon, and the Track 2 and Track 3 crossovers shown here exist only under the assumption that advanced AI discovers a presently unknown class of structural attack. The cases are retained because the consolidated spectrum of Figure 9 includes them, and because ruling primitives out is as much a part of the due-diligence basis of this analysis as identifying exposure; they carry the weakest evidential status of the three tiers defined in Table 1A and Section 4.5, and they inherit the falsifiability conditions of Section 5.

Appendix C.1. SLH-DSA Estimates of CRQC+AI Vulnerability (FIPS 205)

Figures C1A and C1B evaluate the runtime attack and modeled feasibility estimates for breaking SLH-DSA (FIPS 205) under each of the different vulnerability tracks for the four different assumption scenarios.

Note: Hypothesis-only scenario: SLH-DSA contains no lattice structure for the collapse parameter ρ to act on, and its Grover-type exposure is already absorbed into its conservative parameters (Section 4.5). SLH-DSA is assigned zero software coupling and reports no crossover within the modeling horizon; the dated Track 2 and Track 3 figures shown in earlier drafts rested on a hypothesized, presently unsupported novel attack and are withdrawn in favor of the undated hypothesis-only treatment used throughout the body.

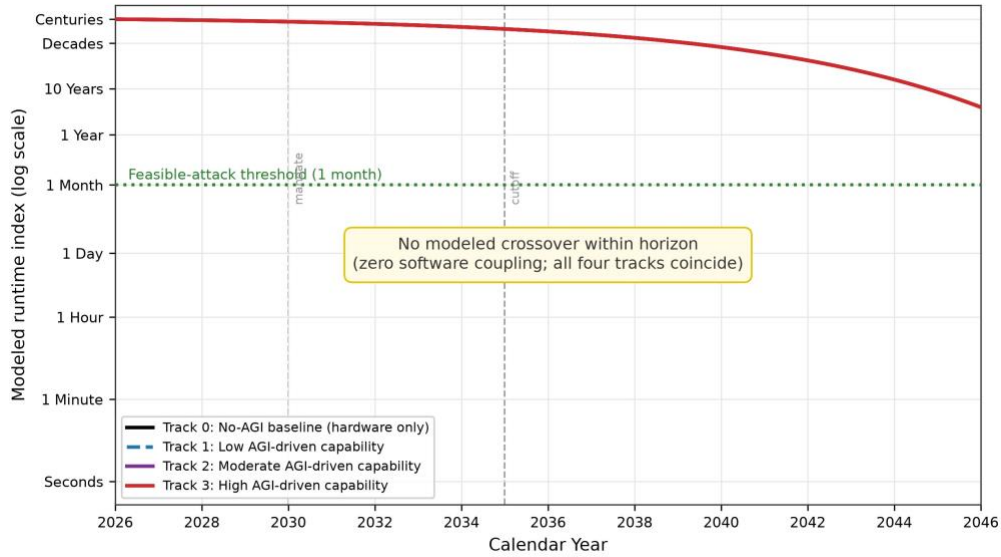


Figure C1A: Projected Attack Runtime Against SLH-DSA FIPS 205
Hash-Based Signatures (conservative)

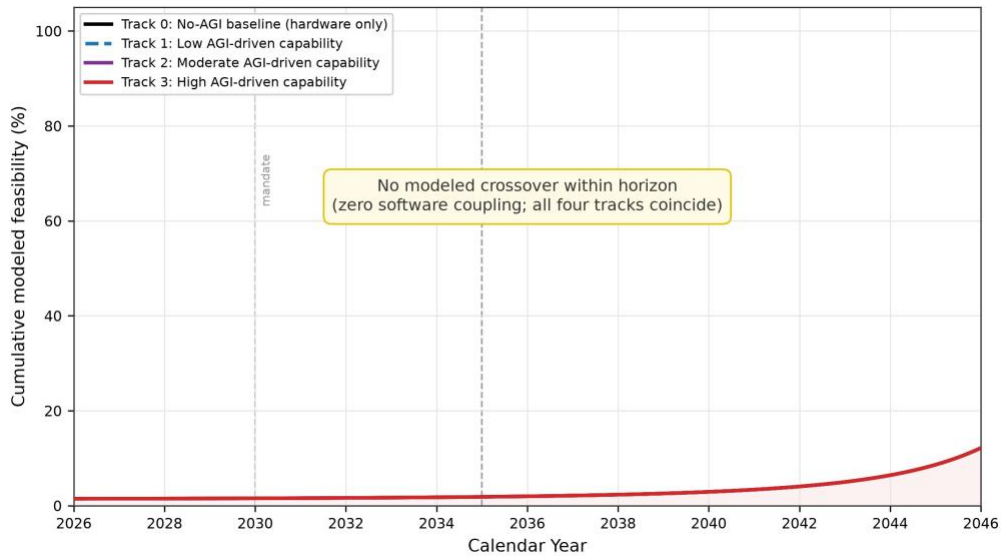


Figure C1B: Modeled Feasibility Estimates for SLH-DSA FIPS 205
Hash-Based Signatures (conservative)

Appendix C.2. AES-256 Estimates of CRQC+AI Vulnerability (FIPS 197)

Figures C2A and C2B evaluate the runtime attack and modeled feasibility estimates for breaking AES-256 (FIPS 197) under each of the different vulnerability tracks for the four different assumption scenarios.

Note: Hypothesis-only scenario: no known quantum, classical, or algorithmic attack breaks AES-256, and under Grover-only analysis it retains 128-bit effective security, so no track shows a feasible break within the horizon. AES-256 is assigned zero software coupling; the dated Track 2

and Track 3 figures shown in earlier drafts derived entirely from speculative assumptions (detailed in the caveat at the end of this section) and are withdrawn in favor of the undated hypothesis-only treatment used throughout the body.

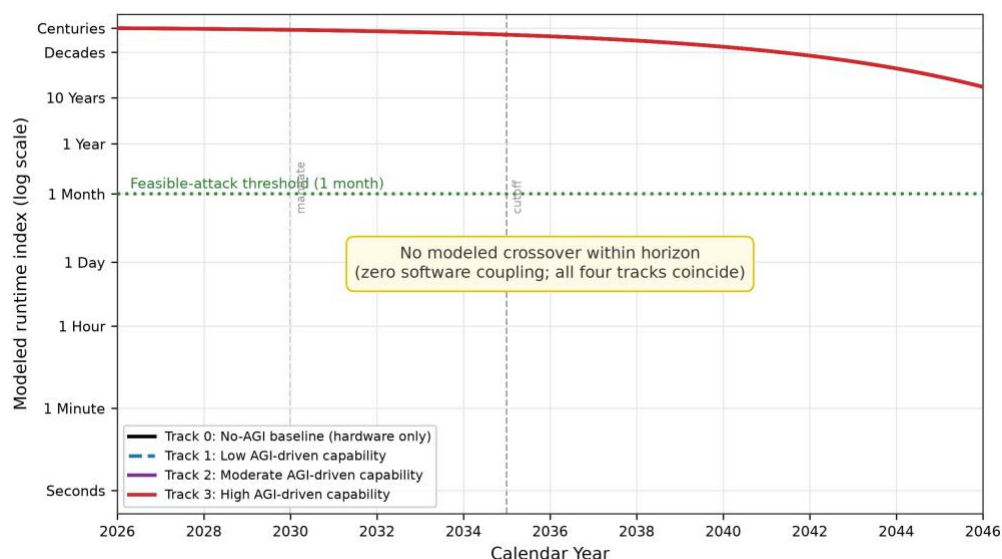


Figure C2A: Projected Attack Runtime Against AES-256 FIPS 197 TLS 1.3 Symmetric Layer,
256-bit (128-bit effective post-Grover)

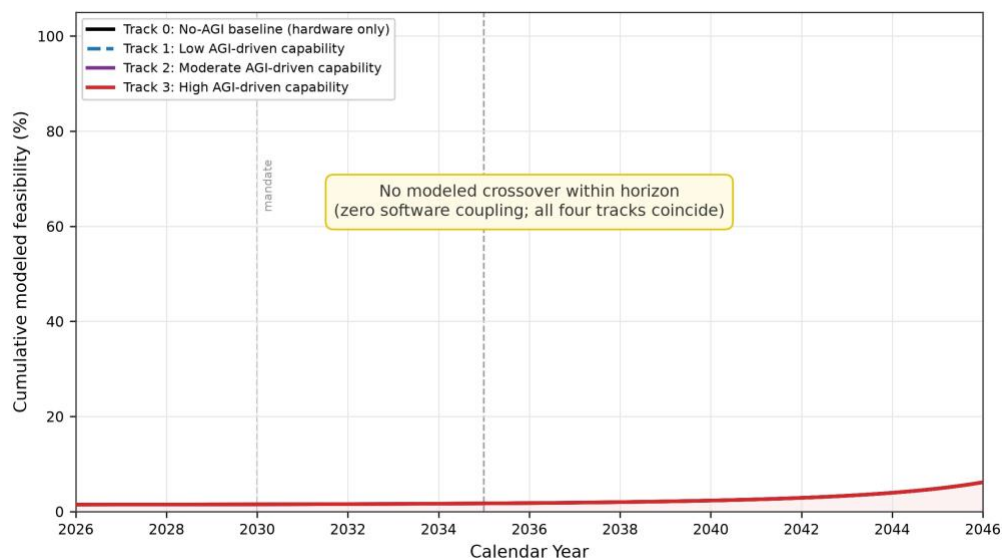


Figure C2B: Modeled Feasibility Estimates for AES-256 FIPS 197 TLS 1.3 Symmetric Layer, 256-bit (128-bit effective post-Grover)

AES, standardized in FIPS 197, remains the relevant approved symmetric cipher for high-security post-quantum deployments, with AES-256 preferred where 128-bit post-Grover security strength is required. Although FIPS 197 includes three different key lengths (128, 192 and 256), the fact that Grover’s algorithm

provides a quadratic (square root) speedup on unstructured search means that the effective quantum security strength for AES is assumed to be one-half of the AES key size. As a result, under the PQC framework advanced by NIST of having encryption algorithms that provide at least 128 bits of quantum security strength, AES-256 is the only one of the three levels of symmetric key length that meets this framework. Accordingly, AES-256 is the only recommended AES symmetric algorithm to be used for PQC encryption CNSA 2.0 and is among the approved security functions listed by NIST. [164]

Quantitatively, Grover’s algorithm searches an unstructured key-space of size 2^k in about $(\pi/4) \cdot 2^{k/2}$ evaluations, so the effective quantum security is half the key length, $k/2$ bits. For AES-256 this maps a classical 2^{256} search to a quantum 2^{128} search, which is why AES-256 alone clears the NIST 128-bit quantum-security bar and underlies the Track 0 and Track 1 baselines in Figures C2A and C2B. This $k/2$ figure is itself a conservative upper bound on the quantum threat. Grover’s search is essentially sequential: the quadratic speedup requires a single coherent computation of depth on the order of $2^{k/2}$, and it parallelizes poorly, so distributing the work across many machines recovers only a square-root benefit per machine. Under the realistic circuit-depth limits that NIST applies, AES-256 retains close to its full classical strength and is placed in the highest post-quantum security category. The Track 2 and Track 3 AES-256 crossovers shown in Figures C2A and C2B therefore do not follow from Grover; they rest entirely on a hypothesized and presently unsupported structural attack and are reported in the hypothesis-only tier defined in Section 4.5.

TLS 1.3 Implication: Under a conventional Grover-only analysis, AES-256 retains 128-bit effective security and is not threatened by brute-force or hardware-scaling attacks. This is reflected in Figures C2A and C2B where Track 0 (baseline) and Track 1 (AGI-enhanced brute force) show no feasible break for AES-256 across the projection horizon. Under the more aggressive AGI-software tracks, however, the updated projections now show AES-256 reaching a feasible-attack threshold late in the horizon, on the order of the mid-2040s under Track 2 (AGI-optimized side-channel and implementation attacks, which are largely independent of key length) and the late-2030s under Track 3 (a hypothesized AGI-discovered structural attack that would defeat the Grover-only bound). AES-256 nonetheless remains the most resistant of the five algorithms evaluated and the last to fall under every Track.

Combined Abstract Figure: Figure C3 as used in the Abstract represents a simplified combination of Figure 9 from the paper and Figures C1A/C1B and C2A/C2B from this appendix.

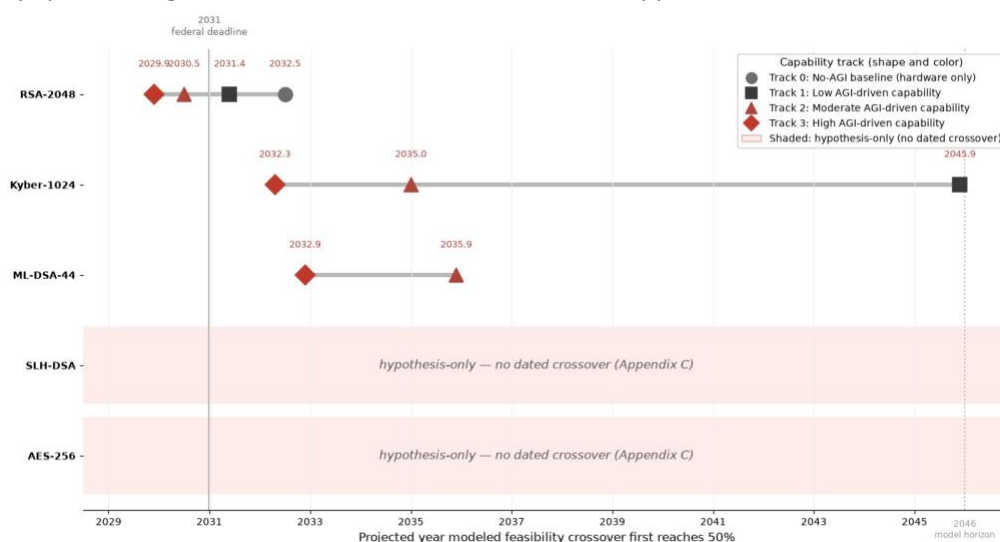


Figure C3: Simplified Combination Risk Spectrum Vulnerability Timelines

Caveat on the AES-256 hypothesis-only treatment: Any dated break estimate for AES-256 would only be a hypothesis-driven scenario projection with no current cryptanalytic basis. No known quantum, classical, or algorithmic attack breaks AES-256, and the consensus view is that it remains secure well beyond the horizon shown. Unlike the RSA and ECC case, where a concrete algorithm (Shor) anchors the projected attack mechanism, the AES-256 curves derive entirely from two speculative assumptions: that an AGI discovers a novel and presently-unknown structural attack on the symmetric cipher, and that side-channel or implementation compromise is treated as equivalent to a break of the cipher itself. The lattice schemes occupy the intermediate, contingency-backed tier: consistent with Section 2.1, HHL and related quantum linear-algebra methods contribute at most polynomial acceleration to an inner step and do not by themselves anchor an attack, so the lattice crossovers are contingent on the conjectured effective-dimension collapse of Section 4.5 rather than on any established algorithm. The AES-256 figures should be read with that limitation expressly in mind.

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